Introduction to Einstein's General Relativity

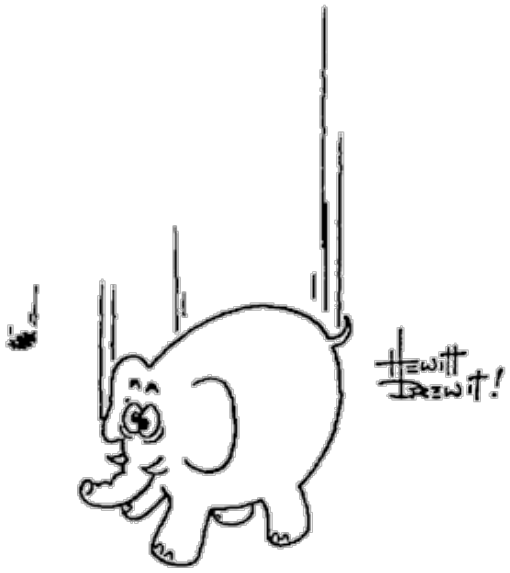
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Abstract

These are Lecture Notes for PHYS 413/513 at Rice University. At the moment, they constitute a work in progress. Please report any typos/errors to mustafa.a.amin@rice.edu.

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Preface

These are notes for a one semester course, PHYS 413/513, at Rice University. The course is meant for students (undergraduate and graduate) who have little or no prior exposure to Einstein's General Relativity. However, the students are expected to have a solid foundation in classical mechanics (with Lagrangians), multivariable calculus, differential equations, and linear algebra. Familiarity with Special Relativity and Electrodynamics will be very helpful too.
Students: Do the recommended reading before class!

Acknowledgements

The primary textbook for the course is James Hartle's "Gravity". The notes rely partly on the textbook for organization, and are very much a work in progress. Don't expect complete notes at the moment. I have added my own examples, explanations and visualizations. where I thought I had something to say which can help students further. I will also supplement these notes with exercises to be done as you read through. This will all happen in due time.

I learnt General Relativity first by reading Sean Carroll's more mathematical textbook, which I thoroughly enjoyed. I was enamoured by the elegant differential geometry at the time, but missed some of the relevant physics in that first pass. Hartle's book helped address that for me. I have also benefited from Misner, Thorne and Wheeler's "Gravitation", Weinberg's presentation in "Gravitation and Cosmology" and Wald's book "General Relativity". More recently, I have enjoyed learning from the Special and General Relativity sections of Thorne and Blandford's "Modern Classical Physics".

I owe an immense gratitude to my teachers – Prof. Eva Silverstein who first asked me to learn GR during a "rotation" during the first year of graduate school, Prof. Renata Kallosh for my first course, and Prof. Robert Wagoner for letting me TA for his GR class. This is how I learnt GR during graduate school. Prof. Barbara Shipman at the University of Texas at Arlington taught me multivariable Calculus and Linear Algebra, and then differential Geometry during my undergraduate years, both in class and often on Friday afternoons beyond class. Her enthusiasm for and pedagogy was infectious, and I hope I can follow in her footsteps.

And, importantly, the previous Phys 413/513 students at Rice were critical in making the course what it is now.

In preparing these notes, I am using some LaTeX and editing assistance from ChatGPT.

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CHAPTER 1

INVITATION

We will explore one of the most impressive achievements of our species. Along the way, I will also tell you how to lose some weight.

1.1 Going along with Gravity

Launch a coconut and a tomato with the same velocity from the earth's surface. In absence of air resistance, they will follow the same trajectory![1] This, in itself is surprising – the trajectory is independent of their mass. If you were launched with them with the same velocity, they would simply appear to float next to you – gravity seems to have disappeared! More generally, if you were falling freely under gravity, by looking at the tomato and coconut (for a short enough period of time) they would each appear to be moving at a constant velocity, with no gravitational acceleration. Even if you, the coconut and tomato were launched with different individual initial velocities, you would still find that they appear to move at a constant velocity relative to you and each other. You will also "feel" weightless.

Exercise 1.1.1 Why is it surprising that in the gravitational field $\mathbf{g}$ the coconut and tomato, which have different masses m_c and m_t, follow the same trajectory? Here $|\mathbf{g}| = |-\nabla\Phi| \approx GM_e/R_e^2$. Consider the situation where the (approximately) constant gravitational field $\mathbf{g}$ near the earth surface is replaced by a constant electric field $\mathbf{E}$, and assume that the fruits have charges q_c and q_t. The accelerations would be $\mathbf{a}_c \propto q_c/m_c$, and similarly $\mathbf{a}_t \propto q_t/m_t$. It depends on the ratio q_i/m_i, ie. on the nature of the bodies, so the trajectories will be different! However, in the case of gravity, $\mathbf{a}_c = \mathbf{a}_t = \mathbf{g}$ independent of the mass! Check these statements using Newton's second law, as well as gravitational and electrostatic forces. See Fig. 1.1

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[1] Check out a version of this experiment on the moon https://www.youtube.com/watch?v=KDp1tiUsZw8. Also see https://www.youtube.com/watch?v=E43-CfukEgs.

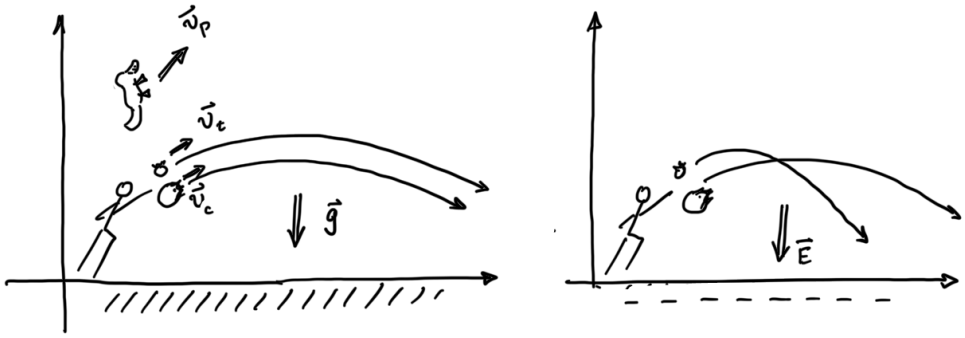


Figure 1.1

Exercise 1.1.2 Suppose that the fruit were launched from the surface of the earth with velocity $\mathbf{v}_\mathrm{c}$ and $\mathbf{v}_\mathrm{t}$. They will follow parabolic trajectories $\boldsymbol{x}_\mathrm{c}(t)$ and $\boldsymbol{x}_\mathrm{t}(t)$, with velocity $\mathbf{v}_\mathrm{c}(t)$ and $\mathbf{v}_\mathrm{t}(t)$ and acceleration $\mathbf{a}_\mathrm{c}(t) = \mathbf{a}_\mathrm{t}(t) = \mathbf{g}$. (i) Write down $\boldsymbol{x}_{\mathrm{c,t}}(t)$ and $\mathbf{v}_{\mathrm{c,t}}(t)$. Now imagine at the same time, a platypus named Polly starts falling freely from some height h, (ii) write down $\boldsymbol{x}_\mathrm{P}(t)$,$\mathbf{v}_\mathrm{P}(t)$ and $\mathbf{a}_\mathrm{P}(t)$. (iii) What would the position, velocity and acceleration of the coconut and tomato be as measured in the "falling frame" of Polly? You should find that $\mathbf{g}$ disappears; Polly will conclude there is no gravity in the situation. (iv) Let the initial velocities $\mathbf{v}_c = \mathbf{v}_P = 0$. The coconut and Polly are falling from a height $z_\mathrm{c}(t = 0) = h_\mathrm{c}$ and $z_\mathrm{P}(t = 0) = h_\mathrm{P}$, both starting at rest. In this case make a sketch of t vs. $z_c(t)$ in the earth's rest frame. Then make the same sketch in Polly's rest frame. What do you notice? Repeat the same exercise with non-zero, and different initial velocities for Polly and the coconut.

While not obvious to us on earth easily (unless you are the free, but unfortunate Polly), get in touch with your favorite "freely falling" astronaut in the international space station to understand how gravitational effects (mostly) disappear for them in their "local" laboratory on the space-station.[2]

[2]In 2024, I read a sign at the Johnson Space Center (inside the mock shuttle), which said astronauts float because of lack of earth's gravity since they are far away from the earth. After this course, you should tell me what you you think about this statement.

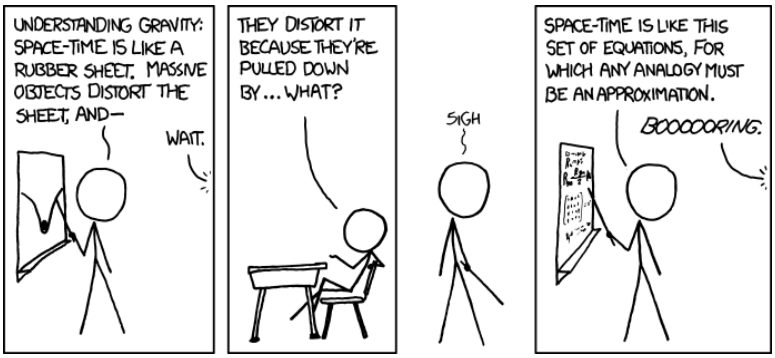


Figure 1.2: Credit: https://xkcd.com

With Einstein as our guide, these simple observations lead us to an astonishing insight into Nature. Gravity is unlike other known forces – it is really about the nature of space and time. Three dimensional space is not a passive arena in which events take place in time, but space and time together – *spacetime* – is an active, dynamical entity. It is a smooth collection of events: a 3+1 dimensional spacetime, whose geometry is determined by the mass-energy within it. The mass of the earth "curves" the spacetime around it.

As we will learn, despite the parabolic locus of the trajectory in space, the coconut and tomato are moving in this curved *spacetime* on the "straightest" possible path which is independent of the composition and mass of the fruit. Gravity is a manifestation of this curved spacetime.

1.2 The road ahead

We will spend much of the course developing a quantitative, calculable and testable understanding of the above ideas. Our underlying theoretical framework for gravity is that of Einstein's General Theory of Relativity (GR). This theory, arguably one of humanity's most elegant and well tested ideas, is built upon Einstein's Special Theory of Relativity. We will review Einstein's Special Theory of Relativity (SR) first – with an emphasis on geometry. The path to GR then is a matter of living with SR "locally" in small enough regions of spacetime, and developing an understanding of how to connect the happenings in many such small regions together. We will compare and contrast our developing understanding of GR with Newtonian ideas of gravity, and see why Newtonian ideas fail to be consistent with SR and more importantly, with observational and experimental tests. Nevertheless, it will be helpful to keep the following connection between Newtonian and Einstein gravity at the back of our mind: $\ddot{\boldsymbol{x}} = -\nabla\Phi \rightarrow$ geodesic equation, $\nabla^2\Phi = 4\pi G\rho \rightarrow$ Einstein Equations.

1.3 Newton vs. Einstein

The Newtonian theory of gravity is of course still extremely useful and accurate for most applications on the earth, solar system and in astrophysics more generally. However, Newtonian gravity fails when considering systems with mass M and extent R, such that GM/Rc^2 becomes order unity. Here $c = 3\times10^{10}\mathrm{cm\,s^{-1}}$ is the speed of light in vacuum and $G = 6.67\times10^{-8}\,\mathrm{dyne\,gm^{-2}\,cm^2}$

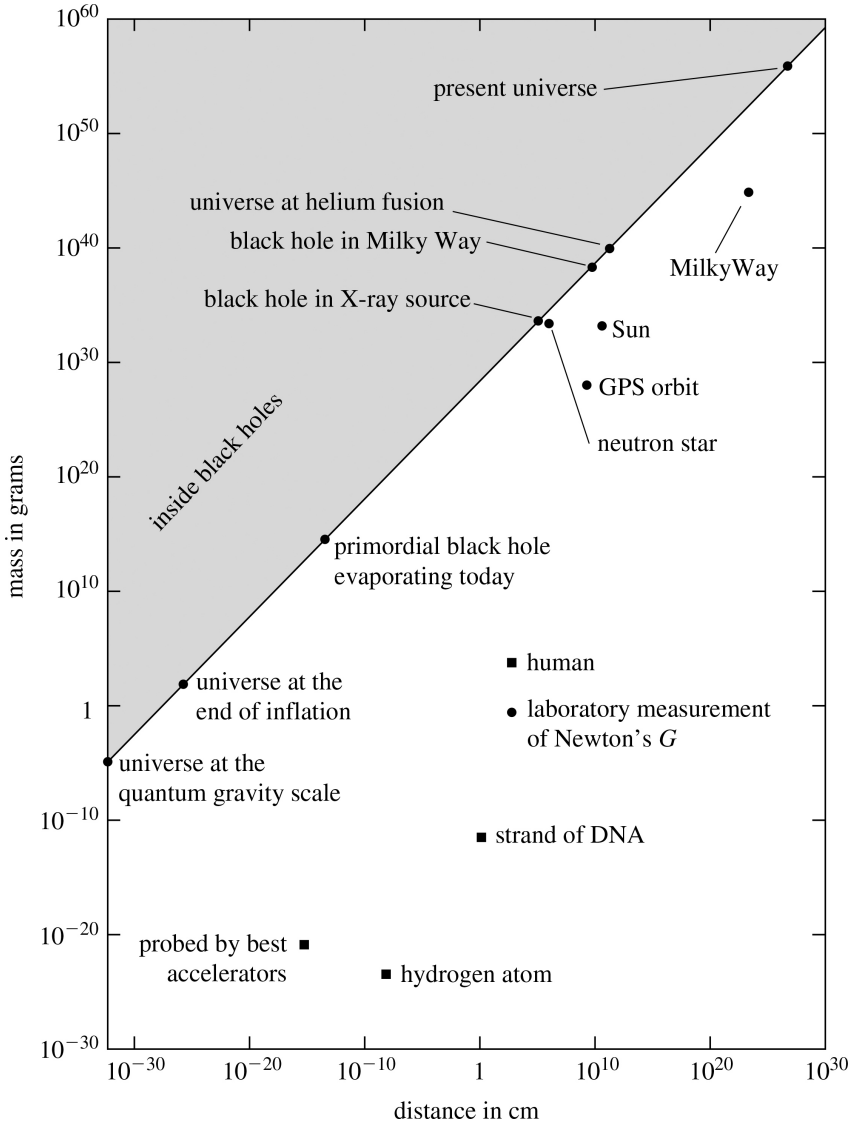


Figure 1.3: General relativity effects are not ignorable near the diagonal line, where $2GM/Rc^2 = 1$. Figure from J. Hartle. To understand the diagonal line, recall that the escape velocity from the surface of a body of mass M and radius R is $v_{\rm esc} = \sqrt{2GM/R}$. When M and R are such that this escape velocity $v_{\rm esc} \to c$, even light cannot escape from its surface (this is the line $M = (c^2/2G)R$). This is the historical definition of a "black hole" (Michell & Laplace, 18th century).

is Newton's gravitational constant. Near a Black Hole, $GM/Rc^2 \sim 1$, neutron star $\sim 10^{-1}$, near the sun (say at the orbital radius of mercury) it is $\sim 10^{-8}$ and $\sim 10^{-6}$ near the sun's surface, whereas near the earth's surface $GM/Rc^2 \sim 10^{-9}$. Note that near the earth and the sun, the precision of the measurements is sufficiently high that even for such small values of GM/Rc^2, we need to take deviations from Newtonian gravity into account. See Fig. 1.3 to get a feel for where different objects lie in the size-mass plane, and where how far they are from the $GM/Rc^2 \sim 1$

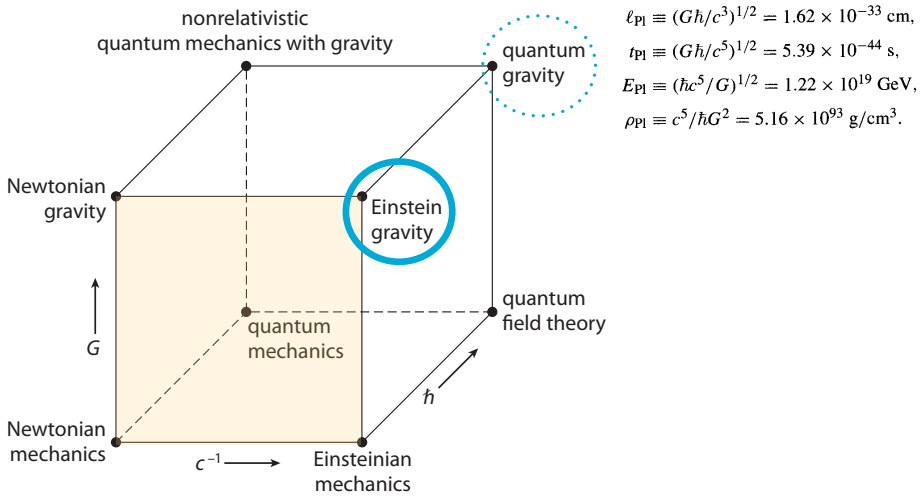


Figure 1.4: Cube of Physics, adapted from A. Zee.

region.

It is very useful to remember that for the sun $GM_\odot/c^2 \sim 1\,\text{km}$, where $M_\odot \sim 2 \times 10^{30}\,\text{kg}$.

Exercise 1.3.1 Estimate $GM/(Rc^2)$ for (i) a proton (ii) human (iii) moon (iv) earth (v) sun (vi) neutron star. Order of magnitude is fine.

1.4 What can GR do for you?

In the course, we will learn to appreciate (aspects of) the working of the Global Positioning System (GPS) relevant for your smart phones to help you orient yourself. Moving further out from our abode on our remarkable third rock from the sun, we will learn how the geometric understanding of spacetime explains the puzzling orbit of Mercury around the sun, and the observed deflection of light by the sun.[3] It also describes why we can occasionally see the same supernova go off multiple times – seeing the "echoes" of the same explosion. We will be able to describe the formation and environment of spectacular ultra-compact "objects", neutron stars and black holes in our cosmos. We will be able to understand how a dance of their binaries generate ripples in our dynamical spacetime, which have now been detected on our rock.[4] Time-permitting we will also touch upon how the evolution of the universe as a whole relies on our understanding of the laws governing a dynamical spacetime.

[3]Light deflection from the sun was first observed during a total solar eclipse of 1919. For a bit of history and its significance, you might like a talk I gave on this in the context of the solar eclipse that was viewable from the continental US in 2024. https://www.youtube.com/watch?v=D1LGSSoLfNg.

[4]See https://www.nobelprize.org/prizes/physics/2017/press-release/.

1.5 Cube of Physics

To conclude, let us also put general relativity in the context of classical and quantum mechanics. A nice visual way is via the "cube of physics", see Fig. 1.4. You should read this as, for example, when G becomes more relevant, then Newtonian gravity plays a prominent role. We will not have much to say about when quantum gravity in this course. It becomes important at the Big Bang and near the center of black holes. Notice the physical scale where quantum gravitational effects need to be taken into account. For this course, we will restrict ourselves to classical gravity.

CHAPTER 2

GEOMETRY AS PHYSICS

Gravity and Geometry Considerations of universal free-fall, amongst other things, eventually led Einstein to the profound insight that gravity is not a force; it is a manifestation of the *geometry of spacetime*. He was helped by the fact that his Special Theory of Relativity (ignoring gravitational effects), initially formulated in terms of transformation laws between different observers, had already been geometrized by Minkowski.[1] This geometry, whether in Special or General Relativity, is $3+1$ dimensional (3 spatial and 1 temporal dimension). In General Relativity, spacetime can be intrinsically curved, and this curvature encodes gravitational effects.

Geometry of space and spacetime is something accessible to experimental verification. Gauss could in principle have tested whether the angles of a large triangle sum to π (the triangle is in three dimensions, not restricted to be on the earth's surface).[2] Modern cosmological observations perform the same kind of experiment on vastly larger scales by comparing known physical scales with their observed angular sizes. You can learn about Gauss's failed attempt to do so in the vicinity of earth. And then learn about our very successful attempt to do so in the context of cosmology. Both are described in Chapter 2 of Hartle.

2.1 Euclidean vs. intrinsically curved spatial geometry

To understand "intrinsically curved" geometry, let us first restrict ourselves to spatial geometries (as opposed to spacetime geometry). To make things easier still, let us consider 2 dimensional spatial geometries.

2.1.1 Two dimensions

2d Euclidean An example of a Euclidean geometry is a 2-dimensional plane, where Euclid's axioms are satisfied. For example, (i) parallel lines do not intersect, (ii) sums of interior angles of

[1] Einstein initially resisted Minkowski's geometrization of Special Relativity, reportedly dismissing it as "superfluous learnedness." He is also quoted as saying, "Since the mathematicians have invaded the theory of relativity, I do not understand it myself anymore." The geometric viewpoint later became indispensable to Einstein in developing general relativity.

[2] This is historically debated a bit.

a triangle made of straight lines is π radians and (iii) the ratio of the circumference of a circle to its radius is 2π. Non-Euclidean geometry is of course a geometry where any of Euclid's axioms are violated.

2d intrinsically curved An example of an intrinsically curved 2-dimensional geometry is a 2-sphere (with radius a). Note that Euclid's axioms are not satisfied on a 2-sphere. For example: (i) Any two distinct great circles intersect, so there is no global analogue of distinct parallel straight lines on the sphere. (ii) Sums of interior angles of a triangle made of straight lines (great circles) is $> \pi$ radians ($= \pi + A/a^2$). Here, A is the area of the triangle on the 2-sphere. (iii) The ratio of the circumference of a circle to its radius is $< 2\pi$. That is, circumference(C)/radius (r)$= 2\pi \sin(r/a)/(r/a)$ where r is the radius of the circle on the 2-sphere.

Note that these properties can be tested by a bug living on the surface of the sphere, without ever leaving the surface. These are " intrinsic" properties of the surface, not requiring any information beyond the surface itself. For example, a bug could take a small segment of a string, and attach one end at some point on the sphere. Then crawl away till the string becomes taut. The bug would measure this taut length in terms of their own tiny steps. This is a radius r in units of their step size. Now, the bug can walk around keeping the string taut to trace out a circle on the sphere. Again, the circumference C can be measured in terms of the number of their tiny steps. By comparing the circumference to the radius, the bug would discover that they were living in a non-Euclidean space. They never had to see the embedded 2-sphere in 3d Euclidean space by flying out of the surface! Also note they can look for these effects without knowledge of global properties such as whether their space is finite or infinite.

A geometry is *intrinsically curved* if departures from Euclidean geometry can be detected by measurements made entirely within the space, even locally, without reference to an embedding in a higher-dimensional space. The opposite is *intrinsically flat*. Thus, a sphere is intrinsically curved, while a plane is intrinsically flat. A cylinder provides an instructive example: although it looks curved when embedded in three dimensions, it is intrinsically flat.

Intrinsically curved spatial geometries are necessarily non-Euclidean, although the term "non-Euclidean" by itself can also include global differences that do not involve intrinsic curvature.[3]

2.1.2 Three dimensions

3d Euclidean Our usual Euclidean 3d space is the obvious generalization of a plane from 2 to 3d.

3d intrinsically curved A 3-sphere is a 3d generalization of the 2-sphere. It is also non-Euclidean. For example the volume enclosed by a 2-sphere of radius r in this 3-sphere geometry is

$$V = 2\pi a^3 \left[\frac{r}{a} - \sin\left(\frac{r}{a}\right)\cos\left(\frac{r}{a}\right)\right]. \tag{2.1.1}$$

In the limit $r/a \ll 1$, this volume approaches its Euclidean counterpart $V = (4\pi r^3/3)[1 - (1/5)r^2/a^2 + \ldots]$.

[3]Euclidean geometries are intrinsically flat, but the converse need not hold globally. For example, an infinite cylinder is intrinsically flat and locally Euclidean, but is not globally equivalent to the Euclidean plane.

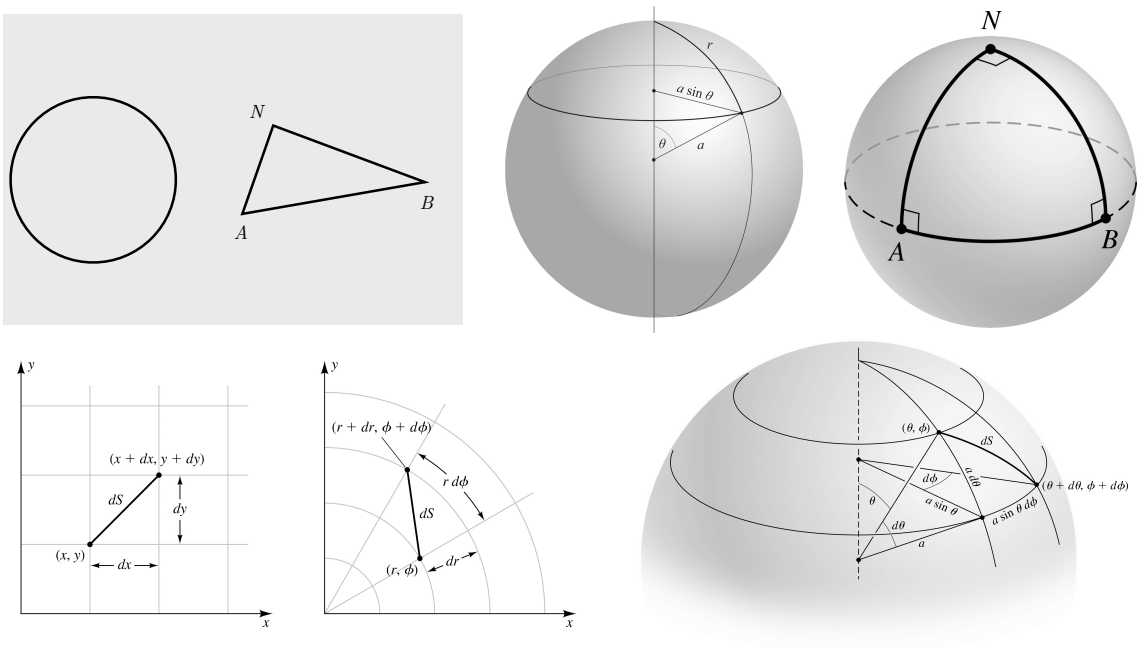


Figure 2.1: Top panel (right): The sum of the interior angles of a triangle ABN on a 2-sphere is $> \pi$, in this particular case it is $3\pi/2$. On the plane it is of course π. Top panel (middle sphere): The circumference of the circle of radius r is $< 2\pi r$, it is $C = 2\pi r \sin(r/a)/(r/a)$. On a plane it would be $C = 2\pi r$. Bottom panel (left): Cartesian and Polar co-ordinates in a 2d plane. Bottom panel (right): r and θ co-ordinates on the sphere. Part of the Fig. adapted from Hartle.

2.2 Specifying a geometry

How does one specify a geometry? A convenient and general way to do this is to define a rule for determining distances between nearby points. This is called a metric. Once this rule is specified, lengths of curves can be calculated by adding up such small distances. "Straight" lines can now be defined as curves with extremal lengths between two points. Circles (or spherical surfaces) can be defined as the locus of all points equidistant from a given point, angles can be defined as ratios of some arclength of a circle to the radius, and so on. This way of defining a geometry has significant advantages over trying to describe a geometry by embedding it in some higher dimensional Euclidean space or specifying it with a set of axioms. Both of these get cumbersome and difficult as we move to general geometries which do not have high degree of symmetry.

If the distances along the surface between all nearby cities of the earth were given, you would have been able to tell that the earth is not a plane. Indeed, if someone told you all the distances between any 4 four non-collinear cities, you would know that the surface of the earth is not planar. Please inform your Flat-Earther buddies.

2.2.1 Labeling and calculating with co-ordinates

Co-ordinates are labels of points in a geometry. For example, in a 2d plane, one can label points using the usual Cartesian co-ordinates (x, y). One could also label points using polar co-ordinates

(r, θ). Different co-ordinates can be chosen to label points in the same geometry. Different co-ordinate systems are connected by a co-ordinate transformation. For example, $x = r\cos\theta$ and $y = r\sin\theta$.

2.3 Spatial metric

The square of the distance, $\mathrm{d}S^2$, between nearby points in a plane with labels (x, y) and $(x + \mathrm{d}x, y + \mathrm{d}y)$ or (r, θ) and $(r + \mathrm{d}r, \theta + \mathrm{d}\theta)$

$$\mathrm{d}S^2 = \mathrm{d}x^2 + \mathrm{d}y^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2. \tag{2.3.1}$$

In different co-ordinate systems the expression for the distance can look different, as can be seen above. Here is the metric in yet another co-ordinate system, $\mathrm{d}S^2 = 2\mathrm{d}\tilde{x}^2 + \mathrm{d}\tilde{y}^2 - 2\mathrm{d}\tilde{x}\mathrm{d}\tilde{y}$. This one also represents a 2d plane. The co-ordinates are linked by $(\tilde{x}, \tilde{y}) = (x, x + y)$. Changing co-ordinate systems does not change the geometry.

Exercise 2.3.1 Using $x = r\cos\theta$ and $y = r\sin\theta$, show that $\mathrm{d}x^2 + \mathrm{d}y^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2$.

For a 3d Euclidean space, the distance between nearby points (x, y, z) and $(x+\mathrm{d}x, y+\mathrm{d}y, z+\mathrm{d}z)$ or equivalently, (r, θ, ϕ) and $(r + \mathrm{d}r, \theta + \mathrm{d}\theta, \phi + \mathrm{d}\phi)$ is

$$\mathrm{d}S^2 = \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 = \mathrm{d}r^2 + r^2(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2). \tag{2.3.2}$$

The different co-ordinate systems are connected by $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$ and $z = r\cos\theta$.

Similarly, for a non-Euclidean geometry like a 2-sphere of radius a, a convenient choice of co-ordinates are the latitudes and longitudes (θ, ϕ). The distance between nearby points (θ, ϕ) and $\theta + \mathrm{d}\theta, \phi + \mathrm{d}\phi)$ is

$$\mathrm{d}S^2 = a^2(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2). \tag{2.3.3}$$

For a 3-sphere of radius a, the distance between nearby points (ψ, θ, ϕ) and $(\psi+\mathrm{d}\psi, \theta+\mathrm{d}\theta, \phi+\mathrm{d}\phi)$ is

$$\mathrm{d}S^2 = a^2[\mathrm{d}\psi^2 + \sin^2\psi(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2)]. \tag{2.3.4}$$

While not necessary, you can embed this 3-sphere ib 4d Euclidean space as a 3-surface $x^2 + y^2 + z^2 + w^2 = a^2$.

Note that this square of small distances can be conveniently written as

$$\mathrm{d}S^2 = \sum_{i=1}^{3}\sum_{j=1}^{3} g_{ij}\mathrm{d}x^i\mathrm{d}x^j, \tag{2.3.5}$$

where $x^1 = \psi$, $x^2 = \theta$ and $x^3 = \phi$, where $g_{11} = a^2$, $g_{22} = a^2\sin^2\psi$, $g_{33} = a^2\sin^2\psi\sin^2\theta$ with all other $g_{ij} = 0$ (the off-diagonal terms need not always be zero). We will use a simplified notation

$$\mathrm{d}S^2 = g_{ij}\mathrm{d}x^i\mathrm{d}x^j, \tag{2.3.6}$$

where summation over repeated upstairs and down stairs indices is implied.[4] Note that g_{ij} is a function of the co-ordinates $g_{ij}(x^1, x^2, x^3)$, which we will write as $g_{ij}(x^k)$ in shorthand.

For a given co-ordinate system, you can think of g_{ij} as components of a $d \times d$ matrix $\boldsymbol{g}$ where d is the dimension of space, with g_{ij} being the entry on the i-th row and j-th column. These entries will depend on the co-ordinate system. For example, in the case of the metric in 3d Euclidean space, the matrix entries in Cartesian co-ordinates would be $g_{ij} = \delta_{ij}$ where $\delta_{ij} = 1$ for $i = j$ and 0 for $i \neq j$. In Polar co-ordinates, $g_{11} = 1$, $g_{22} = r^2$ and $g_{33} = r^2 \sin^2\theta$, and all other entries being 0. The components (matrix entries) look different, but together with $\mathrm{d}x^i \mathrm{d}x^j$ in the respective co-ordinates, they must yield the same $\mathrm{d}S^2$.

In general, the infinitesimal, $\mathrm{d}S^2 = g_{ij}\mathrm{d}x^i\mathrm{d}x^j$, that determines the distance between any two nearby points is called the metric. In the case, since we are restricting ourselves to space and not spacetime, it is the spatial metric. It is also commom practice to call g_{ij} the metric also.

Exercise 2.3.2 (i) Write down the components g_{ij} on a 2-sphere of radius a in terms of θ and ϕ co-ordinates. (ii) Consider the "Pinched Peanut" geometry defined by $\mathrm{d}S^2 = a^2(\mathrm{d}\theta^2 + f^2(\theta)\mathrm{d}\phi^2)$ where $f(\theta) = \sin\theta(1 - \sin^2\theta)$. Write down the components g_{ij} in terms of θ and ϕ co-ordinates.

2.3.1 Path length and volumes

Path length : To calculate non-infinitesimal distances along some curve γ, we can use the metric (in any co-ordinate system) as follows:

$$S[\gamma] = \int_\gamma \mathrm{d}S = \int_{\lambda_1}^{\lambda_2} \mathrm{d}\lambda \sqrt{g_{ij}\frac{\mathrm{d}x^i}{\mathrm{d}\lambda}\frac{\mathrm{d}x^j}{\mathrm{d}\lambda}}. \tag{2.3.7}$$

In the final expression, you should think of a parametrized curve $x^i(\lambda)$, and $g_{ij}(x^k(\lambda))$. The length of the curve is of course independent of the choice of co-ordinates, so it can be computed in any co-ordinate system.

Suppose Bob the bug is walking on a 2-sphere, leaving a localized chemical trail. Will the length of this chemical trail depend on the co-ordinates or parametrization we use to characterize the curve. Of course not! Nevertheless, depending on the geometry of space and the curve in question, it is useful to make a good co-ordinate choice for computational ease. It might be also useful sometimes to use one of the appropriately chosen co-ordinates itself as the parameter λ.

Important: Geometric quantities, for example, the distance between two nearby points, or the length of a particular curve, does not depend on the choice of co-ordinate systems or parametrization.

Exercise 2.3.3 Consider a 2d plane. A parabola passing through the origin can be parametrized as $x^1(\lambda) = \lambda^2/2$ and $x^2(\lambda) = \lambda$, where x^1 and x^2 are Cartesian co-ordinates. Write down g_{ij} and

[4] Notice the potentially confusing notation, $\mathrm{d}x^2 \neq (\mathrm{d}x)^2$ in these later expressions, although earlier we were using $\mathrm{d}x^2 = (\mathrm{d}x)^2$.

$\mathrm{d}x^i(\lambda)/\mathrm{d}\lambda$. Compute the length between $\lambda = 0$ and $\lambda = 1$. Hint: You might find the following substitution $\lambda = \sinh u$ useful for doing the integration.

Volumes To calculate the volume of a spatial region enclosed by a surface $\mathcal{S}$

$$\mathcal{V} = \int_{\mathcal{V}} \sqrt{\det g} \prod_{i=1}^{d} \mathrm{d}x^i, \tag{2.3.8}$$

where $\det g$ is the determinant of $\boldsymbol{g}$. To get some intuition, consider a metric which is diagonal in some co-ordinate system ($g_{ij} = 0$ for $i \neq j$). Then the infinitesimal distance along each co-ordinate direction when the co-ordinates change by $\mathrm{d}x^i$ is $\mathrm{d}S_i = \sqrt{g_{ii}}\mathrm{d}x^i$ (no summation). And the volume will be $\prod_{i=1}^{d} \mathrm{d}S_i$. Note that the diagonality of g_{ij} means the co-ordinate curve intersect at right angles.

Exercise 2.3.4 Use the metric on the surface of a 2-sphere to do the following: (i) Calculate the circumference a circle of radius r (distance from the center to the circumference) in the two-dimensional geometry given by the surface of a sphere of radius a. Show that this reduces to $2\pi r$ when $r \ll a$. (ii) Calculate the area of the same circle on the sphere, and again show that it reduces to πr^2 when $r \ll a$. Bob, the bug lives here. (iii) Do the same for a cylinder of radius a and infinite extent in the perpendicular direction to the radius. Assume $r < \pi a$. Note that the cylinder is not globally Euclidean, but it is intrinsically flat.

Exercise 2.3.5 Beetle Bina lives in a different two-dimensional geometry with metric

$$\mathrm{d}S^2 = \mathrm{d}r^2 + a^2 \sinh^2(r/a)\, \mathrm{d}\phi^2. \tag{2.3.9}$$

Here $0 \leq \phi < 2\pi$, and r is the intrinsic radial distance from $r = 0$. (i) Calculate the circumference $C(r)$ of a circle of radius r. (ii) Calculate the area $A(r)$ enclosed by this circle. (iii) Show that for $r \ll a$,

$$C(r) = 2\pi r \left[1 + \frac{r^2}{6a^2} + \cdots\right], \tag{2.3.10}$$

and

$$A(r) = \pi r^2 \left[1 + \frac{r^2}{12a^2} + \cdots\right]. \tag{2.3.11}$$

Compare these results with those for a Euclidean plane. For a circle of fixed radius r, does Bina have more area enclosed or does Bob? (See the previous problem for Bob).

Having hopefully gained some intuition about what intrinsically curved means, let me re-iterate that while we started with just curved space for pedagogy, ultimately we care about curved spacetime when thinking about gravitational physics.

For spacetime, the role played by Euclidean spatial geometry in the examples in this chapter, will be taken over by Minkowski spacetime geometry (which is non-Euclidean!).[5] And, we will define instrincally curved spacetime geometry by their departure from Minkowski spacetime geometry.

2.4 Practice Problems and Beyond

1. **Triangles on a Sphere**: Draw the following triangles on a 2-sphere (each side is part of a great circle). (i) The sum of interior angles is almost π. (ii) The sum of interior angles is almost 2π.

2. **Arc Lengths**: Using $\mathrm{d}S = \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2}$ show that the perimeter enclosing a half circle (the arc + straight line) of radius a, $C = \oint \mathrm{d}S = \oint \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} = \pi a + 2a$. Similarly, use $\mathrm{d}S = \sqrt{\mathrm{d}r^2 + r^2\mathrm{d}\theta^2}$ in polar co-ordinates to find C also.

3. **More Peanuts**: Consider the peanut geometry described by the metric $\mathrm{d}S^2 = a^2(\mathrm{d}\theta^2 + f^2(\theta)\mathrm{d}\phi^2)$ where $f(\theta) = \sin\theta(1 - 3\sin^2\theta/4)$. For $\theta = \Theta < \pi/2$ in this geometry, calculate the radius of the cap (you can't leave the surface again) enclosed by the circle defined by $\theta = \Theta$, and how the circumference of this circle is connected to the radius.

4. **Spherical Geometry 2d** Consider a latitude where $\theta = \Theta < \pi/2$ on a 2-sphere of radius a. Show that the radius of the spherical cap (you can't leave the surface!) enclosed by this latitude is $r = a\Theta$, the length of the latitude is $2\pi a \sin\Theta$. A useful generalization, if you want to know the distance between any two points on such a sphere (without restricting to the same latitude or longitude is as follows: The distance between two points on a 2-sphere or radius a, with co-ordinates (θ_1, ϕ_1) and (θ_2, ϕ_2) is

$$S = a\cos^{-1}\left[\sin\theta_1\sin\theta_2 + \cos\theta_1\cos\theta_2\cos(\phi_1 - \phi_2)\right]. \tag{2.4.1}$$

   See, for example, here for details.

5. **Not-so-flat Earth**: If the distances along the surface between all nearby cities of the earth were given, you would have been able to tell that the earth is not a plane. Indeed, if someone told you all the distances between any 4 four non-collinear cities, you would know that the surface of the earth is not planar. Let P_1, P_2, P_3 and P_4 be the four cities, with $S_{ij} = S_{ji}$ being the distance between them (length of the shortest curve in the surface between those points). Then, the 4-points lie in a plane if and only if:

$$\det\begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & S_{12}^2 & S_{13}^2 & S_{14}^2 \\ 1 & S_{21}^2 & 0 & S_{23}^2 & S_{24}^2 \\ 1 & S_{31}^2 & S_{32}^2 & 0 & S_{34}^2 \\ 1 & S_{41}^2 & S_{42}^2 & S_{43}^2 & 0 \end{pmatrix} = 0. \tag{2.4.2}$$

[5] The metric (in spatially Cartesian co-ordinates) is

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2. \tag{2.3.12}$$

It is the negative sign that makes it non-Euclidean.

Check this, and then please inform your flat-earther buddies. If you want to confirm that we are on an (idealized) 2-sphere, then the following must be true:

$$\det \begin{pmatrix} 1 & \cos S_{12} & \cos S_{13} & \cos S_{14} \\ \cos S_{21} & 1 & \cos S_{23} & \cos S_{33} \\ \cos S_{31} & \cos S_{32} & 1 & \cos S_{34} \\ \cos S_{41} & \cos S_{42} & \cos S_{43} & 1 \end{pmatrix} = 0. \tag{2.4.3}$$

For proofs, and more, see this post by Terence Tao here.

2.5 Appendix

2.5.1 Curves of Extremal Length

The expression for $S[\gamma]$ is the length of any given curve γ:

$$S[\gamma] = \int_\gamma \mathrm{d}S = \int_{\lambda_1}^{\lambda_2} \mathrm{d}\lambda \sqrt{g_{ij} \frac{\mathrm{d}x^i}{\mathrm{d}\lambda} \frac{\mathrm{d}x^j}{\mathrm{d}\lambda}}. \tag{2.5.1}$$

Question: Given some points P and Q in our geometry, I want to find the curve connecting those points such that its length is a minimum (actually just "stationary"). The answer is provided by the curve $x^i(\lambda)$ which satisfies the **Euler-Lagrange equations**:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial(\mathrm{d}x^i/\mathrm{d}\lambda)}\right) - \frac{\partial L}{\partial x^i} = 0, \qquad \text{where} \qquad L = \sqrt{g_{ij} \frac{\mathrm{d}x^i}{\mathrm{d}\lambda} \frac{\mathrm{d}x^j}{\mathrm{d}\lambda}}. \tag{2.5.2}$$

Curves satisfying this equation have the property that between two fixed end points, if I try to change the curve γ to $\gamma + \delta\gamma$, the length of the curve does not change at first order in the small changes $\delta S[\gamma] = S[\gamma + \delta\gamma] - S[\gamma] = 0 + \mathcal{O}[\delta\gamma^2]$. The analogues of straight lines in arbitrary spatial geometries are called *geodesics*. They are curves whose length is stationary under sufficiently small variations with fixed endpoints, the ones we just found above.

Straight Line in a plane: Let me show you the use of this "technology" to go through a long route to find the equation of a straight line in a 2d plane. While seemingly trivial, understanding how it is used here will help you gain confidence in using it in more complicated situations later. I will also introduce the (sometimes confusing) notion of an affine parameter.

Consider two points P and Q in a plane. Let us put Cartesian co-ordinates (x, y) on the plane. Consider a collection of curves with P and Q as their endpoints. Each curve is parametrized by λ: $(x(\lambda), y(\lambda))$. We move along a curve from P to Q as λ changes. We wish to find the curve connecting P and Q with extremal length. Of course you intuitively know that this must be a straight line, ie. $\mathrm{d}y/\mathrm{d}x = const.$ We will prove this. In terms of the parameterized curve, you perhaps might also be expecting the equation that the curve satisfies is $\mathrm{d}^2x(\lambda)/\mathrm{d}\lambda^2 = 0$ and $\mathrm{d}^2y(\lambda)/\mathrm{d}\lambda^2 = 0$. We will see that this will not be true in

general for arbitrary choice of λ, but will turn out to be true a special class of them. This special class is: $\lambda = aS + b$ where S is the arc length along the curve for arbitrary choice of $a \neq 0$ and b.

Let us begin. The length of any curve γ from P to Q

$$\begin{aligned} S[\gamma] = \int_\gamma \mathrm{d}S = \int \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} = \int \mathrm{d}\lambda \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\lambda}\right)^2}, \\ = \int \mathrm{d}\lambda\, L, \quad \text{where} \quad L = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\lambda}\right)^2}. \end{aligned} \tag{2.5.3}$$

If a curve extremizes its length, then $\delta S = 0$. Such a curve satisfies the Euler-Lagrange equations

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial(\mathrm{d}x/\mathrm{d}\lambda)}\right) = \frac{\partial L}{\partial x}\,, \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial(\mathrm{d}y/\mathrm{d}\lambda)}\right) = \frac{\partial L}{\partial y}. \tag{2.5.4}$$

These equations yield:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right) = 0\,, \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}y}{\mathrm{d}\lambda}\right) = 0\,. \tag{2.5.5}$$

Since these equations imply $L^{-1}\mathrm{d}x/\mathrm{d}\lambda = const.$ and $L^{-1}\mathrm{d}y/\mathrm{d}\lambda = const.$, taking the ratio of these constants is also a constant. That is $\mathrm{d}y/\mathrm{d}x = const.$ This is of course the expected equation for the locus of a line.

However, these equations do not yield $\mathrm{d}^2x/\mathrm{d}\lambda^2 = 0$ and $\mathrm{d}^2y/\mathrm{d}\lambda^2 = 0$. Now, look back at the equation for S. You will notice that $\mathrm{d}\lambda L = \mathrm{d}S$, where $\mathrm{d}S$ is the infinitesimal length along the curve. So our equation for $x(\lambda)$ can be written as (after multiplying by L^{-1})

$$\frac{1}{L}\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right) = 0 \implies \frac{\mathrm{d}^2x}{\mathrm{d}S^2} = 0, \tag{2.5.6}$$

where we used $L\mathrm{d}\lambda = \mathrm{d}S$. Similarly for the y equation.

Are there a general class of parameters λ' for which $\mathrm{d}^2x/\mathrm{d}\lambda'^2 = 0$? Clearly if $\mathrm{d}S = \mathrm{d}\lambda'/a$ or equivalently, if $\lambda' = aS + b$, then again $\mathrm{d}^2x/\mathrm{d}\lambda'^2 = 0$. Similarly for y. Thus all parameters λ' which are related linearly to the distance along the curve S will yield the desired simple equations. These are called **affine parameters**.

Note that regardless of the parametrization, the locus is $\mathrm{d}y/\mathrm{d}x = const.$ – it is a line as you expect. However $\mathrm{d}^2x/\mathrm{d}\lambda^2 = 0$ and $\mathrm{d}^2y/\mathrm{d}\lambda^2 = 0$ for along this straight line locus of points only if we chose λ to be an affine parameter.

CHAPTER 3

FROM NEWTONIAN SPACE AND TIME TO MINKOWSKI SPACETIME

I am combining aspects of Ch. 3 and Ch. 4 of Hartle here.

3.1 Newtonian Space, Time and Reference Frames

The arena of Newtonian physics is a 3-dimensional Euclidean space and a universal time. However, this seemingly obvious (from everyday experience at least) Newtonian separation into an absolute Euclidean space plus a universal time will be replaced by a "spacetime" by the end of this chapter. To get there, we will first need to understand the notion of inertial reference frames, Galilean and Lorentz transformations between such frames, as well as the Principle of Relativity. I hope to make each step seem obvious, and yet the conclusion is anything but obvious.

Reference Frames In Euclidean space, imagine an idealized 3d rigid lattice of rods. At each intersection there is an idealized clock, with all ticking at some universal rate. I like to think of tiny demons sitting at these intersections, meticulously recording whatever happens in their vicinity. This lattice of rods and clocks defines a *reference frame*. It allows us to label where and when events take place.

Although the lattice naturally suggests Cartesian co-ordinates, we could equally well use spherical polar co-ordinates, or some other choice. How we label points is a choice; the reference frame is the same. Furthermore, not just co-ordinates of a given reference frame, but the choice of reference frame itself does not change the underlying (assumed Euclidean) geometry of space, and universality of time.

Different reference frames can be displaced, rotated, rotating, or moving with respect to each other. Among this large family, there is a special class of frames called *inertial frames*. Within the Newtonian space and time context, we define them below.

3.1.1 Inertial Reference Frames

Go far away from large masses and shield yourself, as well as you can, from other long-range forces. Set up a reference frame with Cartesian co-ordinates (x, y, z) and synchronized clocks. If

a free particle moving through the frame satisfies

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \frac{\mathrm{d}^2z}{\mathrm{d}t^2} = 0, \tag{3.1.1}$$

then your frame is *inertial*. Any frame moving with constant velocity with respect to an inertial frame is also inertial. A rotated (but not rotating) or displaced version of an inertial frame is also inertial.

3.2 Galilean Transformations

Consider two inertial frames $\mathcal{S}$ (Sammy's frame) and $\mathcal{S}'$ (prime is Polly's frame) in the standard configuration: their axes are aligned, their origins coincide at $t = t' = 0$, and $\mathcal{S}'$ moves with speed v along the x-axis of $\mathcal{S}$. Then, the relationships between the co-ordinates and time of an event is give by

$$x = x' + vt', \qquad y = y', \qquad z = z', \qquad t = t'. \tag{3.2.1}$$

For a particle with velocity $(V^{x'}, V^{y'}, V^{z'})$ in $\mathcal{S}'$,

$$(V^x, V^y, V^z) = (V^{x'} + v, V^{y'}, V^{z'}). \tag{3.2.2}$$

Notice that $\mathrm{d}^2x/\mathrm{d}t^2 = 0 \Longrightarrow \mathrm{d}^2x'/\mathrm{d}t'^2 = 0$ and similarly for y' and z'.

3.3 Principle of Relativity

No experiment performed entirely inside an inertial laboratory can distinguish one inertial frame from another.

Consider two inertial frames, which we will imagine to be enclosed opaque boxes (with grids and clocks inside them as before), with our previous $\mathcal{S}$ being Sammy's frame and $\mathcal{S}'$ being Polly's. The Principle of Relativity states that if Sammy and Polly set up exactly the same initial configurations of matter and fields in their respective inertial laboratories, the subsequent evolution of the matter and fields will be identical in their respective frames. That is, the laws of physics are the same in all inertial frames.

Galileo, said it best (from his *Dialogue Concerning the Two Chief World Systems*):

> Shut yourself up with some friend in the main cabin below decks on some large ship, and have with you there some flies, butterflies, and other small flying animals. Have a large bowl of water with some fish in it; hang up a bottle that empties drop by drop into a wide vessel beneath it. With the ship standing still, observe carefully how the little animals fly with equal speed to all sides of the cabin. The fish swim indifferently in all directions; the drops fall into the vessel beneath; [...] When you have observed all these things carefully [...], have the ship proceed with any speed you like, so long as the motion is uniform and not fluctuating this way and that. You will discover not the least change in all the effects named, nor could you tell from any of them whether the ship was moving or standing still. In throwing something to your companion, you will need no more force to get it to him whether he is in the direction of the

> bow or the stern, with yourself situated opposite. [...] the butterflies and flies will continue their flights indifferently toward every side, nor will it ever happen that they are concentrated toward the stern, as if tired out from keeping up with the course of the ship, from which they will have been separated during long intervals by keeping themselves in the air.

Note that Galileo was talking about mechanical phenomenon, our principle, as used by Einstein applies to all phenomenon.

3.3.1 "Light" Trouble for Galilean Transformations

If Sammy and Polly set up identical experiments with a laser in their respective laboratories and measure the speed of light, they both find $c = 3 \times 10^{10}\,\mathrm{cm\,s^{-1}}$. This is consistent with the Principle of Relativity (PoR).

Now let them drill holes in their laboratories so that light emitted by Polly's laser enters Sammy's laboratory. Sammy measures the speed of this incoming light and again finds it to be c.[1] Note that this is a different statement from the previous one: before, Sammy and Polly performed identical experiments within their own laboratories; now they are describing the same physical light sent from Polly's laboratory to Sammy's.

This spells trouble for Galilean Transformations (GT). Suppose Polly's reference frame moves with velocity $v\hat{\boldsymbol{x}}$ relative to Sammy's, and Polly sends the light toward Sammy in the $-\hat{\boldsymbol{x}}$ direction. Polly measures the velocity of the light to be $-c\,\hat{\boldsymbol{x}}$. According to GT, Sammy should therefore measures its velocity to be $(v-c)\hat{\boldsymbol{x}}$, and hence its speed should be $|v-c|$. But Sammy instead measures the speed of the light to be exactly $c = 3 \times 10^{10}\,\mathrm{cm\,s^{-1}}$, independent of the relative velocity v between the two laboratories.

The prediction $|v-c|$ arose directly from GT. If we retain the Principle of Relativity and the observed behavior of light, GT must therefore be wrong.

3.4 Resolution via Lorentz Transformations

We can leave PoR as is, but change GT as follows. Einstein suggested the transformation between frames should be[2]

$$x = \gamma(x' + vt'), \quad y = y', \quad z = z', \quad t = \gamma(t' + vx'/c^2), \tag{3.4.1}$$

where

$$\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{3.4.2}$$

With these transformations, the velocities are related by

$$(V^x, V^y, V^z) = \left(\frac{V^{x'} + v}{1 + vV^{x'}/c^2}, \frac{V^{y'}}{\gamma(1 + vV^{x'}/c^2)}, \frac{V^{z'}}{\gamma(1 + vV^{x'}/c^2)} \right). \tag{3.4.3}$$

[1] This is what Michelson-Morley essentially found, though they were really looking for motion relative to ether.

[2] They carry Lorentz name because he had already discovered these transformations in the context of "ether" theory. Einstein gave them a fundamentally different interpretation.

When $v/c \ll 1$, this reduces to the Galilean result of $V^x \approx V^{x'} + v$ and so on. But if $V^{x'} = \pm c$, then

$$V^x = \pm c, \tag{3.4.4}$$

independent of v.

3.5 The "Price"

We have retained the Principle of Relativity and the observed behavior of light, but replaced Galilean Transformations by Lorentz Transformations. There is a price: the Newtonian notions of a universal time and a frame-independent spatial length can no longer be maintained.

Now? : Recall that under Galilean Transformations, $x = x' + vt', t = t'$. In particular, if two events occur at the same time for Polly, $\mathrm{d}t' = 0$, then they also occur at the same time for Sammy, $\mathrm{d}t = 0$.

Now, recall that with Lorentz transformations $x = \gamma(x' + vt')$ and $t = \gamma\left(t' + vx'/c^2\right)$. For two nearby events,

$$\mathrm{d}t = \gamma\left(\mathrm{d}t' + \frac{v}{c^2}\mathrm{d}x'\right). \tag{3.5.1}$$

Therefore, if Polly regards the two events as simultaneous, $\mathrm{d}t' = 0$, then Sammy finds

$$\mathrm{d}t = \frac{v}{c^2}\mathrm{d}x' \neq 0, \tag{3.5.2}$$

for spatially separated events ($\mathrm{d}x' \neq 0$).

Two events that are simultaneous for Polly need not be simultaneous for Sammy. There is no universal notion of "now."

Length? : This also changes our notion of lengths. Measuring the length of an extended object requires determining the positions of its two ends *simultaneously*, that is, at the same time. Since Sammy and Polly no longer agree on which pairs of events are simultaneous, they need not agree on the spatial length of an object. Spatial separation at a fixed time, like time intervals, can depend on the inertial frame.

3.6 Invariant spacetime interval and Minkowski spacetime

What replaces these separately observer-independent notions of spatial separations and time intervals is a frame-independent quantity associated with *spacetime*. Define

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 \tag{3.6.1}$$

Using Lorentz Transformations, one finds

$$-c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 = -c^2\mathrm{d}t'^2 + \mathrm{d}x'^2 + \mathrm{d}y'^2 + \mathrm{d}z'^2. \tag{3.6.2}$$

Thus, for the same events, Sammy and Polly might find $\mathrm{d}x \neq \mathrm{d}x'$ and $\mathrm{d}t \neq \mathrm{d}t'$, but they agree on the spacetime interval $\mathrm{d}s^2$.

Exercise 3.6.1 For the standard configuration of reference frames, show that $-c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 = -c^2\mathrm{d}t'^2 + \mathrm{d}x'^2 + \mathrm{d}y'^2 + \mathrm{d}z'^2$.

The arena of physics is therefore no longer naturally separated into a universal 3-dimensional Euclidean space, and a universal time. Space and time must instead be regarded as parts of a single 3+1 dimensional spacetime.

Just like the geometry of space was specified by providing $\mathrm{d}S^2$ between any two points in that space, the geometry of the spacetime we have arrived at is defined by the spacetime metric

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2, \tag{3.6.3}$$

where $\mathrm{d}S^2$ is the Euclidean metric.

Here, we have chosen the synchronized time coordinate t of the inertial frame (see below), while allowing arbitrary time-independent coordinates on its Euclidean spatial slices; $\mathrm{d}S^2$ is the Euclidean spatial line element written in whatever spatial coordinates we choose. The geometry defined by $\mathrm{d}s^2$ is the **Minkowski spacetime**. Notice the minus sign. Minkowski spacetime is not Euclidean. But it is not intrinsically curved either.

Inertial Reference frames in Minkowski spacetime We can again imagine a lattice of rods and clocks, with the same procedure of choosing an intertial frame.

The clocks now need to be *synchronized* carefully. A convenient operational prescription is Einstein synchronization: if one clock sends a light signal at t_e, a nearby clock reflects it, and the first receives it at t_r, then the nearby clock is assigned the reflection time $t_{\text{bounce}} = \frac{1}{2}(t_e + t_r)$.

The care is necessary, because if we move clocks from a common location where they would be synchronized, the time they show when they get to their eventual grid point can depend on how we moved them! We will see this by the end of this chapter when we discuss proper time.

If we move from one inertial frame to another, but in each inertial frame we adopt a synchronized time, the (right hand side) of the metric can be written as $-c^2\mathrm{d}t^2 + \mathrm{d}S^2$ where $\mathrm{d}S^2$ is the metric of Euclidean space. We could also label the frame using some co-ordinates that mix space and time, in that case the rhs form will look different.

Exercise 3.6.2 For a given inertial reference frame, consider labeling the Euclidean space with Cartesian co-ordinates, then $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2$. (i) If instead you used polar co-ordinates $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$ and $z = r\cos\theta$ to label the Euclidean space, what would the right hand side be? (ii) Now consider labeling it with $x = \tilde{x}$ and $t = \tilde{t} - \alpha\tilde{x}/c$, $y = \tilde{y}$ and $z = \tilde{z}$ (with $|\alpha| < 1$). What is the form of the metric? Note that this is still the same underlying geometry of Minkowski space.

3.6.1 The Minkowski Metric

In general, the Minkowski metric (for that matter, any spacetime metric) can be written as

$$\mathrm{d}s^2 = g_{\mu\nu}(x^\alpha)\mathrm{d}x^\mu\mathrm{d}x^\nu, \tag{3.6.4}$$

where μ and ν range from 0 to 3 and Einstein summation convention is used over repeated greek indices. Here, $g_{\mu\nu}$ can be thought of as entries of a 4 symmetric matrix. In general its entries can depend on space and time co-ordinates (x^0, x^1, x^2, x^3). We used $g_{\mu\nu}(x^\alpha)$ as a shorthand for the $g_{\mu\nu}(x^0, x^1, x^2, x^3)$. For Minkowski space, if at each fixed-Einstein synchronized-time-slice within an inertial reference frame (with $x^0 = ct$), we choose to label the spatial slice with Cartesian co-ordinates then

$$g_{\mu\nu}(x^\alpha) \equiv \eta_{\mu\nu}, \tag{3.6.5}$$

where

$$\eta_{00} = -1, \quad \eta_{ij} = \delta_{ij}, \quad \eta_{0i} = 0. \tag{3.6.6}$$

When we think of $\eta_{\mu\nu}$ as entries of a matrix $\boldsymbol{\eta}$, then

$$\boldsymbol{\eta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{3.6.7}$$

The entries of the inverse of this matrix are denoted by $\eta^{\mu\nu}$. Note that $\eta^{\mu\alpha}\eta_{\alpha\nu} = \delta^\mu{}_\nu$, where $\delta^0{}_0 = \delta^1{}_1 = \delta^2{}_2 = \delta^3{}_3 = 1$ and 0 for $\mu \neq \nu$. More generally, for any metric, we define $g^{\mu\nu}$ as entries of the inverse of the metric matrix:

$$g^{\mu\alpha}g_{\alpha\nu} = \delta^\mu{}_\nu\,. \tag{3.6.8}$$

Exercise 3.6.3 In an inertial frame in an underlying Minkowski spacetime we choose spacetime co-ordinates such $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2$, where $\mathrm{d}S^2$ is the metric on Euclidean spacetime. If we put spherical polar co-ordinates on the Euclidean space, what are $g_{\mu\nu}$ and $g^{\mu\nu}$ for all μ and ν. Note that μ and ν range from 0 to 3.

3.7 "Moving" in Spacetime

The best way to explore Minkowski spacetime is to think about the paths of particles through it.

For two nearby events:

$$\mathrm{d}s^2 < 0 \qquad \text{timelike,} \tag{3.7.1}$$

$$\mathrm{d}s^2 = 0 \qquad \text{lightlike or null,} \tag{3.7.2}$$

$$\mathrm{d}s^2 > 0 \qquad \text{spacelike.} \tag{3.7.3}$$

The locus of particles (massive or massless) in spacetime are called world lines. Massive particles move on timelike worldlines – that is $\mathrm{d}s^2 < 0$ along every segment of the worldline. On the other hand, light moves on null worldlines with $\mathrm{d}s^2 = 0$.

3.7.1 Spacetime Diagrams

Spacetime diagrams are invaluable for keeping track of events and comparing different frames. Always label the events you are comparing. For simplicity, consider $1+1$ dimensions, and we will be in an inertial frames with co-ordinates chosen so that $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 = -c^2\mathrm{d}t'^2 + \mathrm{d}x'^2$. We draw ct and x on an ordinary sheet of paper, but remember that the geometry of the diagram is not Euclidean.[3] Two distinct events can even satisfy $\mathrm{d}s^2 = 0$, something impossible for Euclidean distance.

Exercise 3.7.1 (i) Draw the wordline of a massive particle at rest $x = x_0$ in the ct vs. x plane. (ii) Draw the wordline of a free massive paricle, and a free massless particle starting at $x = x_0$ at $t = 0$ and moving in the negative x direction. (iii) Draw the wordline of a massive particle undergoing simple harmonic motion around $x = x_1$. (iv) Draw the following event. A very massive particle at rest at $x = 0$ emits two photons in the positive and negative x direction. The photon bounces off mirrors at $x = x_1$ and $x = -x_1$ and return back to the massive particle. Drawn wordlines of all three particles.

3.7.2 Two Frames in the same spacetime diagram

In the standard configuration, for a frame $\mathcal{S}'$ moving at speed v along x relative to $\mathcal{S}$:

Time axis of $\mathcal{S}'$ The origin of $\mathcal{S}'$ satisfies $x' = 0$, so

$$x = vt \quad \text{equivalently} \quad ct = \frac{c}{v}x. \tag{3.7.4}$$

Space axis of $\mathcal{S}'$ Events with $t' = 0$ satisfy

$$ct = \frac{v}{c}x. \tag{3.7.5}$$

These tilted axes already show that simultaneity is frame-dependent. It is very useful to make a grid of constant x' and t' lines in the ct vs. x plane.

Using spacetime diagrams, let us visit time dilation and length contraction:

Time dilation For a clock at rest in $\mathcal{S}'$, $\Delta x' = 0$, so invariance of the interval gives

$$c^2\Delta t'^2 = c^2\Delta t^2 - \Delta x^2. \tag{3.7.6}$$

Since $\Delta x = v\Delta t$,

$$\Delta t = \gamma \Delta t'. \tag{3.7.7}$$

A clock moving relative to us accumulates *less* time between the same two events $\Delta t' < \Delta t$.

[3] This is similar to saying that we can represent the geometry of a 2-sphere (our earth) on a flat sheet using a Mercater projection. The distance between two points on the Mercater projection however is not given by the Euclidean distance between those points!

Length contraction A rod of length L' is at rest in the $\mathcal{S}'$ frame. The length is measure at the same instant of time t' in the $\mathcal{S}'$ frame. We want to know the length in the $\mathcal{S}$ frame, which we can do, by recording the positions of its endpoints at the same time in that frame. The result is

$$L = \frac{L'}{\gamma}. \tag{3.7.8}$$

Be careful: length measurements in two different frames generally involve different pairs of simultaneous events.

Here are some details: In the $\mathcal{S}'$ frame, the end points of the rod are at $x' = 0$ and $x' = L'$. The equation of world line of these endpoints in the ct vs. x plane in the $\mathcal{S}$ frame are $ct = (c/v)x$ and $ct = (c/v)(x - L)$, where L is the length to be determined. Look at where these worldlines curves intersect the $t = 0$ axes, the interval on the x axis is L (we can of course do this at any fixed t, but $t = 0$ is convenient). Using $t = \gamma(t' + vx'/c^2)$, the farther endpoint's t' co-ordinate at $t = 0$, $t' = -L'v/c^2$, which can now be used in $x = \gamma(x' + vt')$ again for the farther endpoint to yield $L = \gamma(L' - v^2L'/c^2) = L'/\gamma$.

Exercise 3.7.2 : (i) Show that for any two timelike separated events $\Delta s^2 < 0$, there exists an intertial frame where the spatial separation $\Delta x = 0$, with $\Delta t \neq 0$. (ii) For two spacelike separated events $\Delta s^2 > 0$, show that there exists an inertial frame where $\Delta t = 0$ and $\Delta x \neq 0$.

3.7.3 Proper Time

For a timelike worldline we define the **proper time**, τ, by

$$\mathrm{d}\tau^2 = -\frac{\mathrm{d}s^2}{c^2}. \tag{3.7.9}$$

Thus

$$\mathrm{d}\tau = \frac{1}{c}\sqrt{-g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu} = \frac{\mathrm{d}\lambda}{c}\sqrt{-g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}}. \tag{3.7.10}$$

In an inertial frame, if we choose co-ordinates so that $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2$, and choose $\mathrm{d}\lambda = \mathrm{d}t$, we get

$$\mathrm{d}\tau = \mathrm{d}t\sqrt{1 - \frac{1}{c^2}\left(\frac{\mathrm{d}\boldsymbol{x}(t)}{\mathrm{d}t}\right)^2}. \tag{3.7.11}$$

If you are momentarily at rest with the particle, then $\mathrm{d}\boldsymbol{x} = 0$, and

$$\mathrm{d}\tau = \mathrm{d}t. \tag{3.7.12}$$

So proper time is simply the time measured by a clock moving along the worldline. It is a geometric, frame-independent quantity.

For a worldline connecting events P and Q, the proper time accumulated along the world line:

$$\tau_{PQ} = \int_P^Q \mathrm{d}\tau. \tag{3.7.13}$$

This is directly analogous to the length of a spatial curve that we discussed in the previous chapter. Another way to think about proper time is the following

It is the time measured on the hand watch of the observer moving on any trajectory (can be an accelerating observer). It measures the length of their worldline, hence it is a frame-independent, geometric quantity.

3.7.4 Free particle in spacetime

In a given inertial frame, what is the equation of motion of a particle moving along a timelike path ($\mathrm{d}s^2 < 0$ everywhere along the path)? For a free particle, we know that this should be a straight line in spacetime line $v(t - t_0) = (x - x_0)$ (ignoring the other two dimensions for simplicity). We can arrive at this known result by extremizing $\tau_{PQ} = \int_P^Q \mathrm{d}\tau$ for a collection of timelike paths passing through the events P and Q. To extremize

$$\tau_{PQ} = \int_P^Q \mathrm{d}\tau = \int_P^Q \sqrt{\mathrm{d}t^2 - \frac{\mathrm{d}x^2}{c^2}} = \int_P^Q \mathrm{d}\lambda \sqrt{\left(\frac{\mathrm{d}t}{d\lambda}\right)^2 - \frac{1}{c^2}\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2} \tag{3.7.14}$$

we are thinking of t and x as functions of some parameter λ, that is a parametrized trajectory $t(\lambda), x(\lambda)$. Then, extremizing the proper time for timelike trajectories would yield the following Euler Lagrange equations:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}t}{\mathrm{d}\lambda}\right) = 0, \quad \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right) = 0, \quad \text{where} \quad L = \sqrt{\left(\frac{\mathrm{d}t}{\mathrm{d}\lambda}\right)^2 - \frac{1}{c^2}\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2} \tag{3.7.15}$$

The EL equations will then yield $L^{-1}\mathrm{d}t/\mathrm{d}\lambda = const.$ and $L^{-1}\mathrm{d}x/\mathrm{d}\lambda = const.$ which implies $\mathrm{d}t/\mathrm{d}x = const.$ This is the equation of a straight line. So extremizing the proper-time gives us a way of getting the equation of motion of a free particle.

If I judiciously choose $\mathrm{d}\lambda = \mathrm{d}\tau$ (remember the straight line case in 2d plane), then we get $\mathrm{d}^2t/\mathrm{d}\tau^2 = \mathrm{d}^2x/\mathrm{d}\tau^2 = 0$. This is a nice simple form for the equation of a straight line in spacetime, it treats space and time symmetrically. We will often parametrize our curves in terms of the proper time along the curve. Note that the extremum is a maximum. Here is something you should keep in mind

For two timelike separated events P after Q, the straight line path provides the maximal proper time.

Why maximal? To understand this, think of an example two events at the same location in space in a given frame. These two can be connected by $c(t_P - t_Q) > 0$. Another trajectory made of two light-like segments moving away from Q and returning at P would yield a zero path length! Note that $\mathrm{d}\tau \leq \mathrm{d}t$. That is, the proper time interval is always shorter than (or equal to) the co-ordinate time interval.[4]

[4]Think about why this is not contradictory to what I said about proper time is maximized on straight line trajectories between two points.

Twin "Paradox" In the often discussed twin paradox (which as we see is not a paradox), it is common to write down the proper time of the traveling twin in terms of the proper time of the stationary one:

$$\tau(t) = \int_0^t d\bar{t}\sqrt{1 - \frac{1}{c^2}\left(\frac{dx}{d\bar{t}}\right)^2}. \tag{3.7.16}$$

Exercise 3.7.3 : Consider twins, one stationary and another with a wanderlust. Suppose the traveling twin travels on a straight line with speed $V = \alpha_1 c$. After a time τ_{turn} as measured in the traveling twin's frame, the twin returns back to at speed $V = \alpha_2 c$. When the twins meet again, the traveling twin has aged by τ. Make a spacetime diagram showing their motion in the frame of the stationary twin. Find the age of the stationary twin at their reunion as a function of α_1, α_2, t_{turn} and τ. [This is a generalization of a Problem 4.9 in Hartle.] Set $\alpha_1 = \alpha_2 = \alpha$, and look at the result, does it make sense? You should note that the contribution to the age of the traveling twin from the infinitesimally small turnaround time is negligible in-spite of the large acceleration needed (see Example 4.3 of Hartle).

The expression for the time t in terms of the proper time of τ is given by

$$t(\tau) = \left[\left\{\int_0^\tau d\bar{\tau}\exp\left[\int_0^{\bar{\tau}} d\tilde{\tau}\frac{a(\tilde{\tau})}{c}\right]\right\}\left\{\int_0^\tau d\bar{\tau}\exp\left[-\int_0^{\bar{\tau}} d\tilde{\tau}\frac{a(\tilde{\tau})}{c}\right]\right\}\right]^{\frac{1}{2}}, \tag{3.7.17}$$

where $a(\tau)$ is the proper acceleration, which can be measured by the traveling twin in their own laboratory with an accelerometer [in the stationary twin's frame, it can be calculated as $\gamma^3 d^2x/dt^2$]. For a derivation of the above formula for $t(\tau)$, see here. Note that this allows the traveling twin to know what the time would be for the stationary one, completely based on their own local measurements (τ and $a(\tau)$).

Exercise 3.7.4 : (i) Use the previous exercise setup but now calculate t for the trip as a function of α and τ using the above expression (assume $\alpha_1 = \alpha_2 = \alpha$). Note that the acceleration is zero everywhere except at turnaround τ_{turn} it can be modeled as a Dirac delta function in $a(\tau) = -c\ln[(1+\alpha)/(1-\alpha)]\delta_D(\tau - \tau_{\text{turn}})$. Note that you could have determined the co-efficient of $\delta_D(\tau - \tau_{\text{turn}})$ given the before and after trajectories with velocities $V = \pm\alpha c$, but I am giving it to you to make your life slightly easier. Hint: Make sure you break up the problem into before and after turnaround when evaluating each $\{\ldots\}$.[5] (ii) This is easier than (i), but do it anyways. Suppose that the traveling twin is on a trajectory with constant proper acceleration $a(\tau) = -g$ which returns them to their stationary twin after a proper time τ. Show that the time that passes for the stationary twin in terms of τ and g is $t = (2c/g)\sinh(g\tau/2c)$.

If you want to understand the twin paradox better, ie. why the traveling twin cannot claim to be stationary and hence reach the conclusion that the other twin must be younger when they

[5] Thanks to Miles for asking a relevant question that made me realize that the problem became harder than it was meant to be.

meet, first thing to realize that one of them is accelerating (which can be measured within their own frame), so they are not equivalent observers. This is true. However, one might argue that we can make the acceleration limited to a very small region of time (ie. think of piecewise two straightline trajectories, or even have a third person take over at the point of turnaround who has always been on the return trajectory). You will find that then the paradox resolves itself when you carefully think about the fact that the traveling twin's perspective does involve change from one inertial frame to another. Clarification of the "paradox" can be found by thinking of surfaces of simultaneity in the different frames. Ask questions like: From the point of view of the traveling twin, just before (one frame of traveling twin) and after (another frame of traveling twin) they change direction, what is the time in the stationary frame? The traveling twin will infer that the twin stationary is younger until the turnaround point. But after turnaround, the traveling twin infers that the stationary twin is older!

If the above word salad is confusing, here is a clean statement. The length of the wordlines of the stationary and traveling twin are both frame-independent, geometric quantities. Done.

3.8 Newtonian Physics as the Low-Speed Limit

For $V \ll c$,

$$\mathrm{d}\tau = \mathrm{d}t\sqrt{1 - \frac{V^2}{c^2}} \simeq \mathrm{d}t\left(1 - \frac{V^2}{2c^2} + \cdots\right). \tag{3.8.1}$$

The relativistic free-particle action

$$S_{\text{free}} = -mc^2 \int \mathrm{d}\tau \tag{3.8.2}$$

therefore becomes

$$S_{\text{free}} \simeq -mc^2 \int \mathrm{d}t + \int \mathrm{d}t\, \frac{mV^2}{2} + \cdots. \tag{3.8.3}$$

Apart from the irrelevant constant term, the familiar Newtonian kinetic action is recovered.

Thus Newtonian physics is not discarded. It is the low-speed limit of the spacetime geometry of Special Relativity.

3.9 Additional Practice Problems

1. Hartle Chapter 3, Problem 4
2. Hartle Chapter 3, Problem 5
3. Hartle Chapter 4, Problem 11
4. Hartle Chapter 4, Problem 14 from Hartle. Use Spacetime Diagrams.

3.10 Appendix

3.10.1 Einstein synchronization and the metric

Why does the metric take the particularly simple form

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2 \tag{3.10.1}$$

when we use the synchronized time of an inertial frame? Consider one spatial direction and allow, for the moment, the more general form

$$\mathrm{d}s^2 = -Ac^2\mathrm{d}t^2 + 2Bc\,\mathrm{d}t\,\mathrm{d}x + C\mathrm{d}x^2. \tag{3.10.2}$$

Einstein synchronization assigns time to clocks using the fact that light has the same one-way speed c in the $+x$ and $-x$ directions. Since light satisfies $\mathrm{d}s^2 = 0$, substituting $\mathrm{d}x = \pm c\mathrm{d}t$ gives

$$-A \pm 2B + C = 0. \tag{3.10.3}$$

The two signs imply $B = 0$ and $C = A$. We furthermore choose t to equal the proper time measured by clocks at rest in the reference frame. For such clocks $\mathrm{d}x = 0$, so $\mathrm{d}s^2 = -c^2\mathrm{d}t^2$, which fixes $A = 1$. Hence

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2. \tag{3.10.4}$$

In three spatial dimensions this becomes

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2, \tag{3.10.5}$$

where $\mathrm{d}S^2$ is the Euclidean spatial metric, written in any time-independent spatial co-ordinates.

Other choices for synchronizing or labeling the clocks describe the same Minkowski spacetime, but the metric need not retain this simple form; in particular, space-time cross terms can appear.

CHAPTER 4

GEOMETRIZING GRAVITY

In the previous lecture we introduced the weak-field, spherically symmetric metric

$$\mathrm{d}s^2 = -\left(1+2\frac{\Phi(x^k)}{c^2}\right)c^2\mathrm{d}t^2+\left(1-2\frac{\Phi(x^k)}{c^2}\right)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j, \qquad \text{with} \qquad \Phi(x^k) = -\frac{GM}{\sqrt{\delta_{ij}x^ix^j}} \ll c^2, \tag{4.0.1}$$

and showed that we recovered the effects of gravitational frequency shifts and time dilation we had argued for based on the equivalence between constant gravitational fields and accelerating frames. We also recovered Newton's law of motion, $d^2x^i/dt^2 = -\delta^{ij}\partial_j\Phi$, for slowly moving particles in gravitational fields. Here $\partial_i \equiv \partial/\partial x^i$.

While this looks like a successful geometrization of gravitational effects, it is not quite complete. For example, if to determine Φ we still use $\nabla^2\Phi(t,\boldsymbol{x}) = 4\pi G\rho(t,\boldsymbol{x})$, where ρ is the mass density, then the theory violates special relativity. A change in ρ somewhere would be communicated to Φ everywhere instantaneously. We will rectify this when we get to Einstein's equations in due time.

Nevertheless, let us investigate this metric and its consequences further. I mentioned that this metric represents a curved spacetime geometry. But how do we know? Perhaps I can define some new co-ordinates $\tilde{x}^\alpha$ such that, in those co-ordinates, the metric becomes $\mathrm{d}s^2 = -c^2\mathrm{d}\tilde{t}^2+\delta_{ij}\mathrm{d}\tilde{x}^i\mathrm{d}\tilde{x}^j$ everywhere. If this were possible, then secretly it was just Minkowski spacetime all along.

To answer this question, we will try to understand the statement of the Equivalence Principle more precisely in the context of the metric (4.0.1). Recall that the Equivalence Principle, written in the context of Newtonian gravity, was:

> *Experiments conducted in sufficiently small freely falling laboratories in a gravitational field will yield the same results as those conducted in inertial laboratories in the absence of gravity.*

Let us expand the gravitational potential about some point $\boldsymbol{x}_0$:

$$\Phi(\boldsymbol{x}) = \Phi(\boldsymbol{x}_0) + \Delta x^i\partial_i\Phi(\boldsymbol{x}_0) + \frac{1}{2}\Delta x^i\Delta x^j\partial_i\partial_j\Phi(\boldsymbol{x}_0) + \ldots, \qquad \Delta x^i = x^i - x^i_0. \tag{4.0.2}$$

For $\boldsymbol{x}_0$ near the surface of the Earth, $\boldsymbol{\nabla}\Phi(\boldsymbol{x}_0) = (GM/R^2)\mathbf{e}_r \equiv g\mathbf{e}_z$, where we have chosen the $+z$ axis to point radially outward. Ignoring the terms quadratic in Δx^i in (4.0.2), and absorbing the constant offset $\Phi(\boldsymbol{x}_0)$ into a constant rescaling of the co-ordinates, we have $\Phi(\boldsymbol{x}) \simeq g\Delta z$. The weak-field metric then becomes

$$\mathrm{d}s^2 \approx -\left(1 + \frac{2g\Delta z}{c^2}\right) c^2\mathrm{d}t^2 + \left(1 - \frac{2g\Delta z}{c^2}\right)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j. \tag{4.0.3}$$

We discussed how, in a sufficiently small freely falling laboratory, the effects of this uniform gravitational field disappear. Within this uniform-field approximation, and to the order in the weak-field expansion in which we are working, it seems plausible therefore that we will find co-ordinates (adapted to the freely falling frame) in which the metric becomes Minkowskian.

This gives us our first important hint: a nonzero $\partial_i g_{00} = -(2/c^2)\partial_i\Phi$ is not by itself evidence that spacetime is curved. Locally, its effects can be removed by choosing co-ordinates adapted to an appropriate freely falling frame. Let us construct such a frame, and co-ordinates adapted to it, more operationally.

Imagine an observer (a free particle, really) in Minkowski spacetime carrying a clock and a laser. They clone themselves and place another freely moving observer nearby, initially at rest with respect to themselves. The first observer sends a pulse of light toward the second observer at time t_e, waits for the reflected pulse to return at time t_r, and defines the radar distance to the second observer by $D = c(t_r - t_e)/2$. They monitor this radar distance and make sure that it does not change—if true, the second observer is part of a developing grid. They repeat the procedure with more cloned observers in three independent spatial directions, and Einstein-synchronize all their clocks. In this way they can construct an inertial reference frame.

Now imagine doing the same thing in the gravitational field around the Earth, but using freely falling observers. You will find that this procedure works for a sufficiently small region of space and for a sufficiently short period of time—like our iPhone drop. The frame so constructed is a *local inertial frame*.

For the uniform-field approximation to our weak-field metric, the co-ordinates adapted to this local inertial frame can be given explicitly. Taking the origin to be the point about which we expanded the potential (4.0.2), so that $\Delta z = z$, define new co-ordinates, retaining terms only to the same weak-field order, by

$$c\tilde{t} = ct\left(1 + \frac{gz}{c^2}\right), \quad \tilde{x} = x\left(1 - \frac{gz}{c^2}\right), \quad \tilde{y} = y\left(1 - \frac{gz}{c^2}\right), \quad \tilde{z} = z + \frac{g}{2c^2}\left(c^2t^2 + x^2 + y^2 - z^2\right). \tag{4.0.4}$$

Notice that the observer at $x = y = z = 0$, who is at rest in the original co-ordinates, has $\tilde{z} = (1/2)gt^2$: they are accelerating upward relative to the freely falling frame. Conversely, an observer at rest at $\tilde{z} = 0$ in the freely falling frame has, to this order, $z \simeq -(1/2)gt^2$, precisely as expected for a freely falling particle. Substituting the co-ordinate transformations gives $\mathrm{d}s^2 = -c^2\mathrm{d}\tilde{t}^{\,2} + \delta_{ij}\mathrm{d}\tilde{x}^i\mathrm{d}\tilde{x}^j$, up to terms beyond the order retained in our weak-field expansion. Thus, to the order in which we are working, the metric has become Minkowskian.

However, if we make the freely falling laboratory larger, or wait long enough, something new happens. Consider two nearby freely falling observers whose separation, measured in the local inertial frame, is $\Delta\tilde{x}^i$. Although the uniform gravitational acceleration has disappeared in this

frame, their relative acceleration does not:

$$\boxed{\frac{\mathrm{d}^2\Delta\tilde{x}^i}{\mathrm{d}\tilde{t}^2} \simeq -\delta^{ij}\partial_j\partial_k\Phi\big|_{\boldsymbol{x}_0}\,\Delta\tilde{x}^k.} \tag{4.0.5}$$

Thus, when $\partial_i\partial_j\Phi \neq 0$, nearby freely falling observers begin to move relative to one another. These are *tidal effects*.

For the uniform-field approximation, $\Phi(\boldsymbol{x}) \simeq \Phi(\boldsymbol{x}_0) + g\Delta z$, we have $\partial_i\partial_j\Phi = 0$, and the freely falling frame behaves like an inertial frame. But for the actual gravitational potential $\Phi(\boldsymbol{x}) = -GM/r$, we have $\partial_i\partial_j\Phi \neq 0$.

This suggests that spacetime curvature should somehow be encoded in second derivatives of the metric, such as $\partial_i\partial_j g_{00} = -(2/c^2)\partial_i\partial_j\Phi$ in our weak-field example. But we should be careful: just as the first derivatives of the metric depend on our choice of co-ordinates, so do its second derivatives. Is there some particular combination of these derivatives that tells us unambiguously whether spacetime is curved?

4.1 The Metric and Local Inertial Frames

Let us now step away from the weak-field approximation and describe spacetime geometry more generally. Recall that an event P in spacetime can be labelled by four co-ordinates $x^\mu(P)$, with $\mu = 0, 1, 2, 3$. Different co-ordinate systems assign different labels to the same events. The geometry itself should of course not depend on how we choose these labels.

For two infinitesimally separated events, with co-ordinate separation $\mathrm{d}x^\mu$, we define the spacetime interval by

$$\boxed{\mathrm{d}s^2 = g_{\mu\nu}(x^\alpha)\mathrm{d}x^\mu\mathrm{d}x^\nu.} \tag{4.1.1}$$

The quantities $g_{\mu\nu}(x^\alpha)$ are the components of the *spacetime metric*. The metric tells us the local geometry of spacetime: given the co-ordinate separation between nearby events, it determines the invariant interval between them.

The interval $\mathrm{d}s^2$ is independent of our choice of co-ordinates, whereas the components $g_{\mu\nu}$ are not. Under a change of co-ordinates $x^\mu \to \tilde{x}^\alpha$, the components of the metric and the co-ordinate differences both change, while $\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu = \tilde{g}_{\alpha\beta}\mathrm{d}\tilde{x}^\alpha\mathrm{d}\tilde{x}^\beta$ remains unchanged.

We have already encountered an example of this. Minkowski spacetime in Cartesian co-ordinates has $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2$, while in spherical polar co-ordinates the same spacetime has $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 + r^2\sin^2\theta\,\mathrm{d}\phi^2$. The metric components in the second set of co-ordinates depend on position, but the spacetime has obviously not become curved simply because we changed our co-ordinates.

Mathematically, how do we figure out whether a given metric represents a curved spacetime? There is a clean answer. We construct from the metric an object called the *Riemann tensor*. Here $g^{\mu\nu}$ denotes the inverse metric, defined by $g^{\mu\alpha}g_{\alpha\nu} = \delta^\mu{}_\nu$. Do not worry for now about where

the following expression comes from; just notice its structure:

$$R_{\alpha\beta\gamma\delta} = \frac{1}{2}\left(\partial_\beta\partial_\gamma g_{\alpha\delta} + \partial_\alpha\partial_\delta g_{\beta\gamma} - \partial_\beta\partial_\delta g_{\alpha\gamma} - \partial_\alpha\partial_\gamma g_{\beta\delta}\right) + \frac{1}{4}g^{\mu\nu}\Big[\left(\partial_\beta g_{\mu\gamma} + \partial_\gamma g_{\mu\beta} - \partial_\mu g_{\beta\gamma}\right)\left(\partial_\alpha g_{\nu\delta} + \partial_\delta g_{\nu\alpha} - \partial_\nu g_{\alpha\delta}\right) - \left(\partial_\beta g_{\mu\delta} + \partial_\delta g_{\mu\beta} - \partial_\mu g_{\beta\delta}\right)\left(\partial_\alpha g_{\nu\gamma} + \partial_\gamma g_{\nu\alpha} - \partial_\nu g_{\alpha\gamma}\right)\Big]. \tag{4.1.2}$$

If $R_{\alpha\beta\gamma\delta} \neq 0$ at a point, spacetime is curved there. We will learn much more about this object later. For now, notice only that it is a particular combination of second derivatives of the metric and terms quadratic in its first derivatives.

We can now state mathematically, and quite generally, what our freely falling construction above accomplished. At any event P, we can choose local inertial co-ordinates X^μ, adapted to a freely falling laboratory passing through P, such that

$$\boxed{g_{\mu\nu}(P) = \eta_{\mu\nu}, \qquad \partial_\alpha g_{\mu\nu}(P) = 0.} \tag{4.1.3}$$

Thus, at P, the metric is Minkowskian and all of its first derivatives vanish. The terms in the Riemann tensor quadratic in first derivatives therefore vanish at P, but the terms involving second derivatives need not. In local inertial co-ordinates the Riemann tensor at P is therefore determined entirely by particular combinations of second derivatives of the metric.

Let us connect this directly to the weak-field metric with which we began. Recall that $g_{00} = -(1+2\Phi/c^2)$. Hence $\partial_i g_{00} = -(2/c^2)\partial_i\Phi$ and $\partial_i\partial_j g_{00} = -(2/c^2)\partial_i\partial_j\Phi$. The first derivatives of the metric therefore contain the familiar gravitational acceleration, whereas the second derivatives contain the tidal field.

Indeed, consider the $0i0j$ components of the Riemann tensor. For a time-independent weak gravitational field, to leading order in Φ/c^2, the terms quadratic in first derivatives of the metric can be neglected, and we obtain

$$\boxed{R_{0i0j} \simeq -\frac{1}{2}\partial_i\partial_j g_{00} = \frac{1}{c^2}\partial_i\partial_j\Phi.} \tag{4.1.4}$$

Thus the relative acceleration of nearby freely falling particles that we found earlier can be written as $d^2\Delta\tilde{x}^i/d\tilde{t}^2 \simeq -c^2\delta^{ij}R_{0j0k}\Delta\tilde{x}^k = -\delta^{ij}\partial_j\partial_k\Phi\,\Delta\tilde{x}^k$.

We can now see geometrically the hierarchy that emerged from our weak-field discussion. The potential Φ appears in the metric. Its first derivatives give the ordinary gravitational acceleration, whose effects can locally be eliminated by going into free fall. Its second derivatives give the tidal field, which cannot in general be eliminated.

The Riemann tensor provides the geometrical description of these tidal effects and hence of spacetime curvature. Its components depend on our choice of co-ordinates, but whether the Riemann tensor vanishes does not. I have not yet given you a good motivation for why the Riemann tensor takes the precise form written above; that will come much later.[1]

[1] Notice an important subtlety here. It is not simply the presence of second derivatives of the metric that tells us that spacetime is curved. The individual quantities $\partial_\alpha\partial_\beta g_{\mu\nu}$ depend on our choice of co-ordinates. It is the particular combination of derivatives appearing in $R_{\alpha\beta\gamma\delta}$ that has geometrical meaning.

We are therefore left with a new question. If spacetime is only locally Minkowskian, what precisely do we mean by a vector, a velocity, or a light cone at an event? To answer this, we first need to understand the local vector space associated with each event in spacetime.

4.2 Spacetime and Tangent Spaces

From the Equivalence Principle to Spacetime Recall the Equivalence Principle: experiments conducted in a sufficiently small freely falling laboratory in a gravitational field give the same results as experiments conducted in an inertial laboratory in the absence of gravity. We made this statement more precise in the previous section. Around any event P, we can choose co-ordinates adapted to a freely falling laboratory such that, at P,

$$\boxed{g_{\mu\nu}(P) = \eta_{\mu\nu}, \qquad \partial_\alpha g_{\mu\nu}(P) = 0.} \tag{4.2.1}$$

Thus, sufficiently locally, spacetime has the familiar geometry of Special Relativity.

What sort of mathematical object allows us to implement this idea at every event, while allowing the geometry to be different from Minkowski spacetime over larger regions?

What is Spacetime? In General Relativity we model spacetime as a *smooth four-dimensional Lorentzian manifold*, denoted by $(\mathcal{M}, g)$. Let us unpack these words.

At the most basic level, $\mathcal{M}$ is a collection of points, which we interpret physically as *events*. We give this collection a topology, which allows us to talk about neighborhoods of events and continuity.[2] We further require that every event P lies in a neighborhood whose events can be labeled continuously by four real numbers $x^\mu = (x^0, x^1, x^2, x^3)$. Such a labeling is called a *co-ordinate chart*. Different charts may be used in different regions of spacetime. Where two charts overlap, we require the transformations between their co-ordinates to be smooth. This makes $\mathcal{M}$ a smooth four-dimensional manifold.[3]

The smooth structure allows us to define what it means for a curve through the manifold to be smooth, independently of the coordinates used to describe it. Now consider all smooth curves passing through some event P. Each curve has a tangent vector at P. The collection of all such tangent vectors forms a four-dimensional real vector space called the *tangent space* at P, denoted by $T_P\mathcal{M}$. At this stage $T_P\mathcal{M}$ is just a vector space. We can add vectors, multiply them by real numbers, and choose bases for the space, but there is not yet a notion of the length of a vector, the angle between two vectors, or whether a vector is timelike, spacelike, or null.

The final ingredient is the *Lorentzian metric* g. At every event P, the metric provides an inner product g_P between vectors in $T_P\mathcal{M}$, and these inner products vary smoothly from event to event. We will usually write this inner product simply as a dot product.

The word "Lorentzian" means that this inner product has signature $(-,+,+,+)$. A basic result of linear algebra then guarantees that at every event P we can choose a basis $\{\boldsymbol{e}_{\hat{0}}, \boldsymbol{e}_{\hat{1}}, \boldsymbol{e}_{\hat{2}}, \boldsymbol{e}_{\hat{3}}\}$

[2] Mathematically this amounts to specifying which subsets of $\mathcal{M}$ are open. We will not need the technical details of the topology.

[3] Notice that the events themselves are not co-ordinates. An event P is a point of $\mathcal{M}$; $x^\mu(P)$ are simply four numbers that we have chosen to label that event.

for which

$$\boxed{\boldsymbol{e}_{\hat{\mu}} \cdot \boldsymbol{e}_{\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}}.} \tag{4.2.2}$$

Such a basis is called an *orthonormal basis*. Thus $T_P\mathcal{M}$, when equipped with the Lorentzian inner product g_P, is a Minkowski vector space.

This statement concerns the vector space at a single event. It does not say that a finite neighborhood of P is Minkowski spacetime. The stronger statement (4.2.1) says that we can choose an actual co-ordinate system in a neighborhood of P whose co-ordinate basis is orthonormal at P, and for which the first derivatives of the metric also vanish there. These are *local inertial co-ordinates*. Physically, they are co-ordinates adapted to a local freely falling frame.

4.3 Vectors and Bases

Co-ordinate Basis Vectors A choice of co-ordinates does more than label events: it naturally provides a basis for the tangent space at every event in the co-ordinate patch.

Consider an event P and the co-ordinate x^μ. Starting at P, imagine varying x^μ while holding the other three co-ordinates fixed. This traces out the x^μ co-ordinate curve through P. The tangent vector to this curve at P defines the co-ordinate basis vector $\boldsymbol{e}_\mu$.

The four co-ordinate curves therefore give four basis vectors $\{\boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$ for $T_P\mathcal{M}$. An infinitesimal displacement from P can then be written

$$\boxed{\mathrm{d}\boldsymbol{s} = \boldsymbol{e}_\mu \, \mathrm{d}x^\mu.} \tag{4.3.1}$$

More generally, any vector $\boldsymbol{V} \in T_P\mathcal{M}$ can be expanded as $\boldsymbol{V} = V^\mu \boldsymbol{e}_\mu$. The four numbers V^μ are the components of $\boldsymbol{V}$ in this co-ordinate basis.

There is an important distinction here. The x^μ are labels assigned to events. The $\boldsymbol{e}_\mu$ are vectors: they live in the tangent space $T_P\mathcal{M}$.

Metric Components The metric tells us the inner products of these co-ordinate basis vectors. Indeed, the components of the metric in a given co-ordinate system are defined by

$$\boxed{g_{\mu\nu} = \boldsymbol{e}_\mu \cdot \boldsymbol{e}_\nu.} \tag{4.3.2}$$

Using (4.3.1), the squared interval is therefore $\mathrm{d}\boldsymbol{s} \cdot \mathrm{d}\boldsymbol{s} = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu = \mathrm{d}s^2$.

The metric components therefore have a very direct geometrical meaning. The diagonal components tell us about the lengths of the co-ordinate basis vectors, while the off-diagonal components tell us about their relative directions. In particular, a co-ordinate basis need not be orthonormal.

For example, consider the Euclidean plane written in polar co-ordinates, for which $\mathrm{d}S^2 = \mathrm{d}r^2 + r^2\mathrm{d}\phi^2$. We therefore have $\boldsymbol{e}_r \cdot \boldsymbol{e}_r = 1$ and $\boldsymbol{e}_\phi \cdot \boldsymbol{e}_\phi = r^2$. Hence $|\boldsymbol{e}_r| = 1$, whereas $|\boldsymbol{e}_\phi| = r$. The latter basis vector is not a unit vector, even though the underlying geometry is simply the flat Euclidean plane.

Co-ordinate versus Orthonormal Bases At any event we are free to choose a basis for $T_P\mathcal{M}$ that is not the basis generated by the co-ordinate curves. Of particular importance is an orthonormal basis, satisfying (4.2.2).

Thus we should distinguish two useful kinds of basis:

- A *co-ordinate basis* $\boldsymbol{e}_\mu$ is generated by the co-ordinate curves and is generally not orthonormal.
- An *orthonormal basis* $\boldsymbol{e}_{\hat{\mu}}$ is chosen so that its inner products are those of Minkowski space, but it need not be a co-ordinate basis.

For example, in the polar co-ordinates above an orthonormal spatial basis is $\boldsymbol{e}_{\hat{r}} = \boldsymbol{e}_r$ and $\boldsymbol{e}_{\hat{\phi}} = \boldsymbol{e}_\phi / r$.

For an observer at P, we can choose $\boldsymbol{e}_{\hat{0}}$ to point along the observer's time direction, while $\boldsymbol{e}_{\hat{1}}, \boldsymbol{e}_{\hat{2}}, \boldsymbol{e}_{\hat{3}}$ define three mutually orthogonal spatial directions in the observer's local laboratory. Components measured in such an orthonormal basis have the same immediate physical interpretation as components measured in an inertial frame in Special Relativity. A local inertial co-ordinate system is more special still: its co-ordinate basis is orthonormal at P, and the first derivatives of the metric vanish there, as in (4.2.1).

Exercise 4.3.1 : Consider a spacetime with metric of the form $\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 + r^2\sin^2\theta\mathrm{d}\phi^2$. For this diagnonal metric, we can define an orthornomal basis set using the co-ordinate basis as follows: $\boldsymbol{e}_{\hat{\mu}} = \boldsymbol{e}_\mu / \sqrt{|g_{\mu\mu}|}$. There is no summation here.

1. What are the components of $\boldsymbol{e}_{\hat{\phi}}$ in the co-ordinate basis? Similary, for $\boldsymbol{e}_{\hat{t}}$, $\boldsymbol{e}_{\hat{r}}$ and $\boldsymbol{e}_{\hat{\theta}}$.
2. What are the components of $\boldsymbol{e}_\phi$ in orthonormal basis? Similary, for $\boldsymbol{e}_t$, $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$.

4.4 Causal Structure

Timelike, Null, and Spacelike Vectors The Lorentzian metric does something qualitatively different from a Euclidean inner product. Since its signature is $(-,+,+,+)$, a nonzero vector $\boldsymbol{V} \in T_P\mathcal{M}$ can have a negative, zero, or positive inner product with itself. We therefore classify vectors according to

$$\boldsymbol{V}\cdot\boldsymbol{V}\begin{cases} < 0, & \boldsymbol{V} \text{ timelike}, \\ = 0, & \boldsymbol{V} \text{ null}, \\ > 0, & \boldsymbol{V} \text{ spacelike}. \end{cases} \tag{4.4.1}$$

This classification is independent of the basis or co-ordinates used to describe the vector, since $\boldsymbol{V}\cdot\boldsymbol{V}$ is a scalar.

To visualize this classification, choose an orthonormal basis at P. Writing $\boldsymbol{V} = V^{\hat{\mu}}\boldsymbol{e}_{\hat{\mu}}$, we have $\boldsymbol{V}\cdot\boldsymbol{V} = -(V^{\hat{0}})^2 + (V^{\hat{1}})^2 + (V^{\hat{2}})^2 + (V^{\hat{3}})^2$. Thus the geometry of vectors at any event is precisely the familiar Minkowski geometry of Special Relativity.

The Light Cone The collection of all null directions in $T_P\mathcal{M}$ forms the *light cone* at P. In an orthonormal basis these directions satisfy

$$\boxed{(V^{\hat{1}})^2 + (V^{\hat{2}})^2 + (V^{\hat{3}})^2 = (V^{\hat{0}})^2.} \tag{4.4.2}$$

The light cone divides the tangent space into timelike directions inside the cone, null directions on the cone, and spacelike directions outside the cone.

Once we choose a time orientation, the timelike and null directions can further be divided into *future-directed* and *past-directed* directions. Thus every event P comes equipped with a local causal structure.

It is worth emphasizing that the light cone at P lives in the tangent space $T_P\mathcal{M}$. In curved spacetime there is a different tangent space, and hence a light cone, at every event.

Curves and Worldlines We can now extend the classification of individual tangent vectors to curves in spacetime. Consider a smooth curve $\gamma(\lambda)$, with tangent vector

$$\boldsymbol{V} = \frac{\mathrm{d}\boldsymbol{s}}{\mathrm{d}\lambda} = \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\boldsymbol{e}_\mu. \tag{4.4.3}$$

A curve is called timelike, null, or spacelike when its tangent vectors are respectively timelike, null, or spacelike along the curve.

The worldline of a physical observer or massive particle is timelike. Light rays have null worldlines. Thus the local light cone determines which directions through an event can be followed by matter or light.

For a timelike worldline, we can use the proper time τ measured by a clock carried along the worldline as its parameter. The tangent vector is then the four-velocity,

$$\boxed{\boldsymbol{u} = \frac{\mathrm{d}\boldsymbol{s}}{\mathrm{d}\tau}, \qquad \boldsymbol{u}\cdot\boldsymbol{u} = -1.} \tag{4.4.4}$$

Thus the four-velocity of any observer points inside the future light cone at each event on their worldline.

For a light ray, $\mathrm{d}s^2 = 0$, and hence no proper time elapses along the null curve. Its tangent vector lies on the light cone.

Causality The light-cone structure tells us which events can potentially influence which other events. Starting from an event P, a physical signal must follow a future-directed timelike or null curve. Events that can be reached from P by such curves lie in the causal future of P; similarly, events from which such curves can reach P lie in its causal past.

Locally, therefore, the Equivalence Principle gives us exactly the causal structure familiar from Special Relativity. The important difference is that in a general spacetime the light cones at different events need not have the same orientation relative to a chosen set of co-ordinates.

Light Cones in Co-ordinates Although the light cone itself is geometrical, its appearance in a co-ordinate diagram depends on the co-ordinates we use. In an orthonormal basis, or in local inertial co-ordinates at P, null vectors have the familiar Special Relativistic form. Thus, if we plot ct vertically and x horizontally with equal scales, the light cone consists of lines at 45°.

In general co-ordinates this need not be true. For example, restricting attention to one spatial co-ordinate x, the metric at an event can have the form

$$\mathrm{d}s^2 = g_{tt}\mathrm{d}t^2 + 2g_{tx}\mathrm{d}t\mathrm{d}x + g_{xx}\mathrm{d}x^2. \tag{4.4.5}$$

The null directions are determined by $\mathrm{d}s^2 = 0$, or equivalently $g_{tt}+2g_{tx}(\mathrm{d}x/\mathrm{d}t)+g_{xx}(\mathrm{d}x/\mathrm{d}t)^2 = 0$. Thus the slopes of the two sides of the light cone in a t–x co-ordinate diagram depend on the metric components.

The two null directions therefore need not appear at 45°. They may appear narrower or wider than 45°, and if $g_{tx} \neq 0$ the cone can also appear tilted. None of this means that the locally measured speed of light differs from c. It reflects the fact that the co-ordinate basis $\{\boldsymbol{e}_t, \boldsymbol{e}_x\}$ is generally not an orthonormal basis.

At any event P, we can instead choose local inertial co-ordinates. Then $g_{\mu\nu}(P) = \eta_{\mu\nu}$, and the light cone at P takes its familiar Special Relativistic form.

4.5 Geodesics

Geodesic equation for a massive particle

The worldline of a freely falling massive particle is such that it makes the proper time between two fixed events stationary. Setting $c = 1$, we therefore consider

$$\tau_{AB} = \int \mathrm{d}\tau = \int \sqrt{-\mathrm{d}s^2} = \int \sqrt{-g_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\beta}{\mathrm{d}\lambda}}\,\mathrm{d}\lambda \equiv \int L\,\mathrm{d}\lambda, \tag{4.5.1}$$

where λ is a parameter along the worldline, and

$$L \equiv \sqrt{-g_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\beta}{\mathrm{d}\lambda}}. \tag{4.5.2}$$

Notice that $L\,\mathrm{d}\lambda = \mathrm{d}\tau$.

The condition $\delta\tau_{AB} = 0$ gives the Euler–Lagrange equations

$$-\frac{\mathrm{d}}{\mathrm{d}\lambda}\left[\frac{\partial L}{\partial(\mathrm{d}x^\sigma/\mathrm{d}\lambda)}\right] + \frac{\partial L}{\partial x^\sigma} = 0. \tag{4.5.3}$$

The Lagrangian depends on $\mathrm{d}x^\sigma/\mathrm{d}\lambda$ explicitly, and on x^σ through the metric $g_{\alpha\beta}(x)$. Its derivatives are

$$\frac{\partial L}{\partial(\mathrm{d}x^\sigma/\mathrm{d}\lambda)} = -\frac{1}{L}g_{\alpha\sigma}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}, \qquad \frac{\partial L}{\partial x^\sigma} = -\frac{1}{2L}\frac{\partial g_{\alpha\beta}}{\partial x^\sigma}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\beta}{\mathrm{d}\lambda}. \tag{4.5.4}$$

Substituting these expressions into the Euler–Lagrange equations gives

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left[\frac{1}{L}g_{\alpha\sigma}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}\right] = \frac{1}{2L}\frac{\partial g_{\alpha\beta}}{\partial x^\sigma}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\beta}{\mathrm{d}\lambda}. \tag{4.5.5}$$

We now use proper time τ as the parameter along the worldline. Since $L\,\mathrm{d}\lambda = \mathrm{d}\tau$, defining $u^\alpha \equiv \mathrm{d}x^\alpha/\mathrm{d}\tau$ turns Eq. (4.5.5) into

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left[g_{\alpha\sigma}u^\alpha\right] = \frac{1}{2}\partial_\sigma g_{\alpha\beta}u^\alpha u^\beta, \tag{4.5.6}$$

where $\partial_\sigma \equiv \partial/\partial x^\sigma$. Expanding the derivative on the left-hand side,

$$g_{\alpha\sigma}\frac{\mathrm{d}u^\alpha}{\mathrm{d}\tau} + \partial_\gamma g_{\alpha\sigma}u^\gamma u^\alpha = \frac{1}{2}\partial_\sigma g_{\alpha\beta}u^\alpha u^\beta. \tag{4.5.7}$$

Since $u^\alpha u^\beta$ is symmetric under $\alpha \leftrightarrow \beta$, we can write

$$\partial_\gamma g_{\alpha\sigma}u^\gamma u^\alpha = \frac{1}{2}\left(\partial_\alpha g_{\sigma\beta} + \partial_\beta g_{\sigma\alpha}\right)u^\alpha u^\beta. \tag{4.5.8}$$

Hence

$$g_{\alpha\sigma}\frac{\mathrm{d}u^\alpha}{\mathrm{d}\tau} + \frac{1}{2}\left[\partial_\alpha g_{\sigma\beta} + \partial_\beta g_{\sigma\alpha} - \partial_\sigma g_{\alpha\beta}\right]u^\alpha u^\beta = 0. \tag{4.5.9}$$

Multiplying by the inverse metric $g^{\sigma\mu}$, and using $g^{\sigma\mu}g_{\sigma\alpha} = \delta^\mu_\alpha$, we obtain

$$\boxed{\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} + \Gamma^\mu_{\alpha\beta}u^\alpha u^\beta = 0, \qquad \Gamma^\mu_{\alpha\beta} \equiv \frac{1}{2}g^{\mu\sigma}\left[\partial_\alpha g_{\sigma\beta} + \partial_\beta g_{\sigma\alpha} - \partial_\sigma g_{\alpha\beta}\right].} \tag{4.5.10}$$

Equivalently,

$$\boxed{\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau} = 0.} \tag{4.5.11}$$

This is the geodesic equation for a freely falling massive particle.

4.6 Review Problems covered in class on Friday Oct 9

1. Work in units where $c = 1$, draw spacetime diagrams as much as possible, it will help!

   Polly is stationary at the origin of an inertial coordinate system (t, x), with line element

   $$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}x^2. \tag{4.6.1}$$

   Sammy is attached to a spring and undergoes simple harmonic motion along the x-axis, following the trajectory

   $$x_s(t) = \frac{v}{\omega}\sin(\omega t) + x_0, \qquad 0 < v < 1, \quad \omega > 0. \tag{4.6.2}$$

   (a) **Accelerometer.** Sammy carries an accelerometer. Find the magnitude of the proper acceleration that it measures as a function of the coordinate time t of Polly's rest frame. For $v \ll 1$, do you recover usual Newtonian results?

   (b) **Light signals.** For this part, assume $x_0 > v/\omega$, so Sammy always remains to Polly's right. Sammy emits two successive light signals toward Polly, separated by a small proper-time interval $d\tau_s$ on Sammy's clock. If the first signal is emitted at coordinate time t, find the interval $d\tau_p$ between the signals' reception by Polly. Express your answer in terms of t and $d\tau_s$.

   (c) **Aging.** For this part, set $x_0 = 0$ and assume $v \ll 1$. Polly and Sammy coincide at $t = 0$ and again after n complete oscillations. Find the difference in their elapsed proper times, through order v^2. Who is older?

   (d) **Orthonormal basis.** At an arbitrary coordinate time t, find the timelike and spacelike orthonormal basis vectors adapted to Sammy's instantaneous rest frame, expressed in terms of the coordinate basis vectors $\boldsymbol{e}_t$ and $\boldsymbol{e}_x$.

   (e) **Energy and momentum.** Return to $x_0 > v/\omega$. At $t = 0$, Polly launches a particle of rest mass m from $x = 0$ toward Sammy. The particle subsequently moves with constant coordinate velocity $q > 0$ (with $|q| < 1$). Suppose the parameters are chosen so that the particle first encounters Sammy at the phase

   $$\omega t_c = \pi. \tag{4.6.3}$$

      i. Determine the required value of q.

      ii. Find the energy of the particle measured by Sammy immediately before the encounter.

      iii. Find the signed spatial momentum of the particle measured by Sammy immediately before the encounter, with positive momentum defined along Sammy's x basis vector.

2. **Co-ordinate and orthonormal basis vectors for diagonal metrics**: Consider a spacetime with metric of the form $\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 + r^2\sin^2\theta\mathrm{d}\phi^2$. For this diagnonal metric, we can define an orthornomal basis set using the co-ordinate basis as follows: $\boldsymbol{e}_{\hat{\mu}} = \boldsymbol{e}_\mu/\sqrt{|g_{\mu\mu}|}$. There is no summation here.

(a) What are the components of $\boldsymbol{e}_{\hat{\phi}}$ in the orthonormal basis? Similary, for $\boldsymbol{e}_{\hat{t}}$, $\boldsymbol{e}_{\hat{r}}$ and $\boldsymbol{e}_{\hat{\theta}}$.

(b) What are the components of $\boldsymbol{e}_{\phi}$ in the co-ordinate basis? Similary, for $\boldsymbol{e}_{t}$, $\boldsymbol{e}_{r}$ and $\boldsymbol{e}_{\theta}$.

(c) What are the components of $\boldsymbol{e}_{\hat{\phi}}$ in the co-ordinate basis? Similary, for $\boldsymbol{e}_{\hat{t}}$, $\boldsymbol{e}_{\hat{r}}$ and $\boldsymbol{e}_{\hat{\theta}}$.

(d) What are the components of $\boldsymbol{e}_{\phi}$ in orthonormal basis? Similary, for $\boldsymbol{e}_{t}$, $\boldsymbol{e}_{r}$ and $\boldsymbol{e}_{\theta}$.

CHAPTER 5

LIGHT RAYS AND THE SCHWARZSCHILD HORIZON

In this chapter, our goal is to understand the nature of the Schwarzschild horizon and the singularity at the center. We begin with light propagation in Minkowski spacetime before turning to Schwarzschild spacetime.

5.1 Light rays in Minkowski spacetime

We begin in flat spacetime, using spherical polar co-ordinates (t, r, θ, ϕ). The line element is

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\Omega^2, \qquad \mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2. \tag{5.1.1}$$

For a radial light ray, $\mathrm{d}\Omega = 0$ and $\mathrm{d}s^2 = 0$, so

$$\frac{\mathrm{d}t}{\mathrm{d}r} = \pm 1, \qquad t - t_0 = \pm(r - r_0). \tag{5.1.2}$$

In the (t, r) diagrams, time is plotted vertically and r horizontally.[1] Consequently the visible slopes are $\mathrm{d}t/\mathrm{d}r = \pm 1$. The null condition specifies two directions through each event, but does not distinguish the future from the past. With the usual time orientation, future-directed light has $\mathrm{d}t > 0$.

Consider a family of inertial observers at rest in these co-ordinates. Their worldlines have $\mathrm{d}r = \mathrm{d}\Omega = 0$, and Eq. (5.1.1) gives

$$\mathrm{d}\tau = \mathrm{d}t. \tag{5.1.3}$$

Thus the co-ordinate time interval Δt is also the elapsed proper time of each stationary inertial observer.

[1] The radial co-ordinate r is nonnegative; an inward ray that passes through the spatial origin subsequently moves outward in the opposite angular direction.

5.2 Radial light rays in Schwarzschild co-ordinates

Outside a spherical, nonrotating gravitating body of mass M, the Schwarzschild line element is

$$\boxed{\mathrm{d}s^2 = -f(r)\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2\mathrm{d}\Omega^2, \qquad f(r) = 1 - \frac{2M}{r}.} \tag{5.2.1}$$

Recall that co-ordinate r need not equal the proper radial distance from the center. For $r > 2M$, we have $f > 0$, and t is a time co-ordinate. The standard future orientation in this exterior region is $\mathrm{d}t > 0$.

For radial light, $\mathrm{d}s^2 = 0$ and $\mathrm{d}\Omega = 0$, giving

$$0 = -f(r)\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f(r)} \qquad \Longrightarrow \qquad \frac{\mathrm{d}t}{\mathrm{d}r} = \pm\frac{1}{f(r)}. \tag{5.2.2}$$

The plus sign is outward-directed and the minus sign inward-directed in the exterior. As $r \to 2M^+$, both co-ordinate velocities, $\mathrm{d}r/\mathrm{d}t = \pm f(r)$, tend to zero. Equivalently, the plotted slopes $\mathrm{d}t/\mathrm{d}r = \pm 1/f(r)$ become increasingly steep.

An ingoing ray sent from the exterior reaches $r = 2M^+$ only at $t \to +\infty$ in Schwarzschild co-ordinates. Similarly, an outgoing exterior ray, when followed backward in Schwarzschild time, approaches $r = 2M^+$ at $t \to -\infty$. Are these divergences properties of the Schwarzschild co-ordinate system? Can we define a better co-ordinate system that helps us better understand the behavior of light near $r = 2M$?

Let us understand if anything interesting is happening to ingoing and outgoing light rays for r near $2M$ and but with $r > 2M$. For an observer at rest (at fixed r), has a 4-velocity $\boldsymbol{u}_{\text{rest}} = \boldsymbol{e}_t/\sqrt{f(r)}$. The orthonormal frame adapted to this observer has basis vectors $\boldsymbol{e}_{\hat{0}} = \boldsymbol{u}_{\text{rest}}$ and $\boldsymbol{e}_{\hat{1}} = \sqrt{f(r)}\boldsymbol{e}_r$. For a ingoing and outgoing light rays, their tangent vectors can be written as $\boldsymbol{k}_\pm = E_{\text{rest}}(\boldsymbol{e}_{\hat{0}} \pm \boldsymbol{e}_{\hat{1}})$ where E_{rest} is the energy measure by that observer. Then, the radial speed of light measured this observer is $-(\boldsymbol{e}_{\hat{1}} \cdot \boldsymbol{k}_\pm)/(\boldsymbol{e}_{\hat{0}} \cdot \boldsymbol{k}_\pm) = \pm 1$. This remains true no matter how close we get to the $r = 2M$, so nothing interesting is happening as far as this observer is concerned.[2]

So perhaps it is a fault of the co-ordinates chosen. We will try to construct a different set of co-ordinates below, where the behavior the behavior of light rays appears more regular.

5.3 Freely falling clocks and Painlevé–Gullstrand (PG) time

Consider radial freely falling observers released from rest at infinity ($e = 1$). Let τ denote proper time along their worldlines. Then their four velocity

$$\boldsymbol{u}_{\text{ff}} = \frac{1}{f(r)}\boldsymbol{e}_t - v(r)\boldsymbol{e}_r, \tag{5.3.1}$$

where we defined

$$v(r) \equiv \sqrt{\frac{2M}{r}}, \qquad f(r) = 1 - v^2(r). \tag{5.3.2}$$

[2]However, note that while the observer measures light moving at speed one, maintaining a stationary position requires an acceleration that diverges as $r \to 2M^+$!

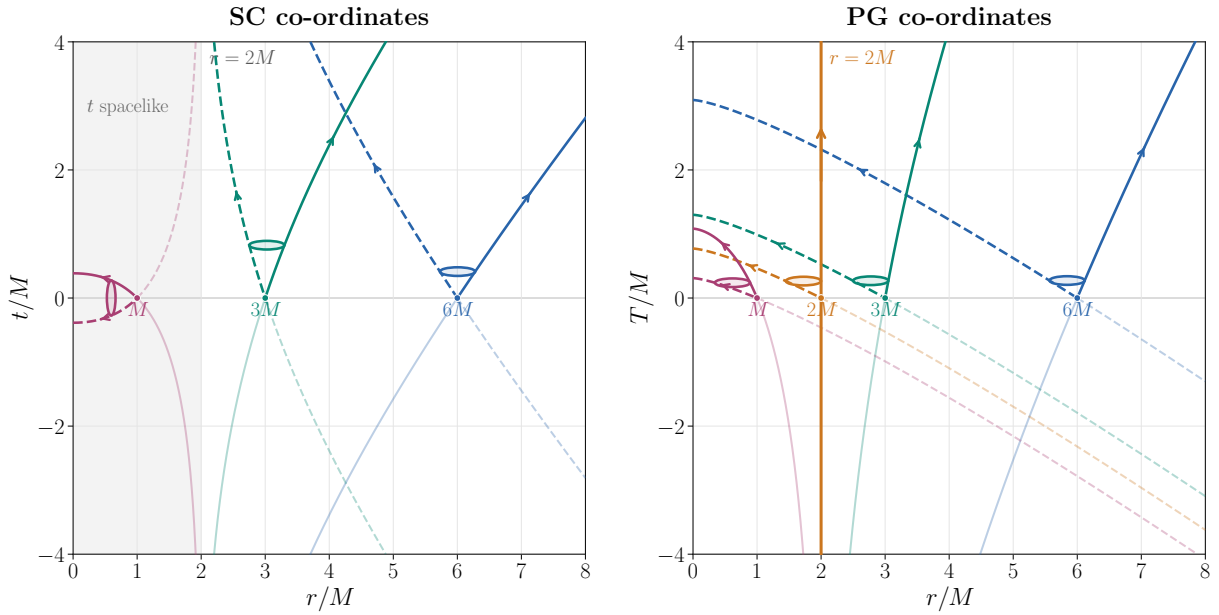


Figure 5.1: Radial null rays in Schwarzschild co-ordinates (left) and ingoing PG co-ordinates (right), from emission radii $6M$, $3M$, $2M$, and M. Solid curves are locally outward null branches and dashed curves locally inward branches. Arrows point to the future; pale extensions indicate the past. Schwarzschild co-ordinates do not assign a regular emission event to $r = 2M$, whereas PG co-ordinates cover the horizon smoothly.

A useful new time co-ordinate T is defined by requiring its constant-time surfaces to be orthogonal to the freely falling four-velocity. In differential form,

$$\mathrm{d}T = -\boldsymbol{u}_{\mathrm{ff}} \cdot \mathrm{d}\boldsymbol{s}. \tag{5.3.3}$$

Along a freely falling observer's worldline, $\mathrm{d}\boldsymbol{s} = \boldsymbol{u}_{\mathrm{ff}}\,\mathrm{d}\tau$, so $\mathrm{d}T = \mathrm{d}\tau$. The new time is therefore the synchronized proper time of such a collection of observers. Using the Schwarzschild metric to evaluate the dot product,

$$\mathrm{d}T = \mathrm{d}t + \frac{v(r)}{f(r)}\,\mathrm{d}r. \tag{5.3.4}$$

Substituting $\mathrm{d}t = \mathrm{d}T - (v/f)\mathrm{d}r$ into Eq. (5.2.1), we find

$$\boxed{\mathrm{d}s^2 = -\mathrm{d}T^2 + (\mathrm{d}r + v(r)\mathrm{d}T)^2 + r^2\mathrm{d}\Omega^2.} \tag{5.3.5}$$

This is the *ingoing Painlevé–Gullstrand* (PG) form of the Schwarzschild metric. Its coefficients are finite at $r = 2M$. Although the transformation from t to T becomes singular there, the PG metric itself does not.

On a constant-T hypersurface, $\mathrm{d}T = 0$, and the spatial metric reduces to

$$\mathrm{d}S^2 = \mathrm{d}r^2 + r^2\mathrm{d}\Omega^2. \tag{5.3.6}$$

The spatial geometry of each PG slice is exactly Euclidean. This does *not* mean that the four-dimensional spacetime is flat! Another way of defining the new time co-ordinate would have been to start with $\mathrm{d}T = \mathrm{d}t + g'(r)\mathrm{d}r$, and now find $g(r)$ such when $\mathrm{d}t = \mathrm{d}T - g'(r)\mathrm{d}r$ is plugged into the $\mathrm{d}s^2$, we find that the $\mathrm{d}T = 0$ slices are spatially flat.

5.4 Radial light rays across the horizon

For radial null propagation, Eq. (5.3.5) becomes

$$0 = -\mathrm{d}T^2 + (\mathrm{d}r + v\mathrm{d}T)^2, \qquad \Longrightarrow \qquad \boxed{\frac{\mathrm{d}T}{\mathrm{d}r} = \frac{1}{-v(r) \pm 1}.} \tag{5.4.1}$$

The plus sign labels light moving locally outward relative to the freely falling observers, and the minus sign labels light moving locally inward. The PG co-ordinate T increases along every future-directed causal worldline, both outside and inside the horizon.

At the horizon the outward-directed null ray has $\mathrm{d}r/\mathrm{d}T = 0$ (equivalenty $\mathrm{d}T/\mathrm{d}r \to \infty$) and remains at $r = 2M$. The inward-directed ray crosses the horizon with $\mathrm{d}T/\mathrm{d}r = -1/2$, without any singularity. Inside $r = 2M$, even the locally outward-directed light moves toward smaller r.

Figure 5.1 compares radial null rays in Schwarzschild and PG co-ordinates. The emission radii are $r_0 = 6M, 3M, 2M, M$. Note that in Schwarzschild co-ordinates the interior and exterior regions separated by $r = 2M$ must be treated as separate patches. In PG co-ordinates all four emission events are regular. The light rays in the two co-ordinate systems themselves are geometrical curves; their appearances differ because their co-ordinate labels differ.

Exercise 5.4.1 Assume a Schwarzschild spacetime with $f(r) = 1-2M/r$ and $v(r) = \sqrt{1 - f(r)} = \sqrt{2M/r}$ below.

(a) **Co-ordinate basis vectors.** Let $\{\boldsymbol{e}_t^{\mathrm{SC}}, \boldsymbol{e}_r^{\mathrm{SC}}\}$ and $\{\boldsymbol{e}_T^{\mathrm{PG}}, \boldsymbol{e}_r^{\mathrm{PG}}\}$ be the co-ordinate basis vectors in the Schwarzschild (SC) and ingoing Painlevé–Gullstrand (PG) co-ordinate systems for the SC spacetime.

Using the co-ordinate transformation $\mathrm{d}T = \mathrm{d}t + v(r)/f(r)\mathrm{d}r$ show that,

$$\boldsymbol{e}_T^{\mathrm{PG}} = \boldsymbol{e}_t^{\mathrm{SC}}, \qquad \boldsymbol{e}_r^{\mathrm{PG}} = \boldsymbol{e}_r^{\mathrm{SC}} - \frac{v(r)}{f(r)}\boldsymbol{e}_t^{\mathrm{SC}}. \tag{5.4.2}$$

It might be useful to note that in general $\boldsymbol{e}_{\beta'} = (\partial x^\alpha/\partial x^{\beta'})\boldsymbol{e}_\alpha$. For your purposes, the unprime and primed notation correspond to the SC and PG co-ordinates respectively. Make sure you recall the notion that when taking a partial derivative with respect to some co-ordinate, all others are held fixed.

(b) **An orthonormal basis.** Using the metric expressed in PG co-ordinates, calculate the three dot products

$$\boldsymbol{e}_T^{\mathrm{PG}} \cdot \boldsymbol{e}_T^{\mathrm{PG}}, \qquad \boldsymbol{e}_T^{\mathrm{PG}} \cdot \boldsymbol{e}_r^{\mathrm{PG}}, \qquad \boldsymbol{e}_r^{\mathrm{PG}} \cdot \boldsymbol{e}_r^{\mathrm{PG}}. \tag{5.4.3}$$

Note that these bases are not orthogonal. For $r > 2M$, $r = 2M$ and $r < 2M$, comment on whether $\boldsymbol{e}_T^{\rm PG}$ and $\boldsymbol{e}_r^{\rm PG}$ are spacelike, timelike, null or not defined.

Construct an orthonormal radial basis adapted to the observers freely falling from rest at infinity. Show that it can be chosen as

$$\boldsymbol{e}_{\hat{0}} = \boldsymbol{e}_T^{\rm PG} - v(r)\boldsymbol{e}_r^{\rm PG}, \qquad \boldsymbol{e}_{\hat{1}} = \boldsymbol{e}_r^{\rm PG}. \tag{5.4.4}$$

Verify that

$$\boldsymbol{e}_{\hat{0}} \cdot \boldsymbol{e}_{\hat{0}} = -1, \qquad \boldsymbol{e}_{\hat{1}} \cdot \boldsymbol{e}_{\hat{1}} = 1, \qquad \boldsymbol{e}_{\hat{0}} \cdot \boldsymbol{e}_{\hat{1}} = 0. \tag{5.4.5}$$

Express both vectors $\boldsymbol{e}_{\hat{0}}$ and $\boldsymbol{e}_{\hat{1}}$ in the SC co-ordinate basis.

Note $\boldsymbol{e}_{\hat{0}}$ is the 4-velocity of a freely falling observer $\boldsymbol{u}_{\rm ff} = \boldsymbol{e}_t^{\rm SC}/f(r) - v(r)\boldsymbol{e}_r^{\rm SC}$ (although the rhs is ill defined in SC co-ordinates at $r = 2M$, the lhs is a perfectly sensible vector). This is perfectly well-defined inside and outside the horizon. By contrast, observers at rest in PG co-ordinates have a 4-velocity $\boldsymbol{u}_{\rm rest} = \boldsymbol{e}_T/\sqrt{f(r)}$, such observers cannot exist for $r < 2M$. For $r \leq 2M$, there is no way to stay at fixed r!

(c) **Radial light rays.** Let $\boldsymbol{k}$ be the future-directed tangent vector to a radial light ray, and let

$$E_{\rm ff} = -\boldsymbol{k} \cdot \boldsymbol{u}_{\rm ff} = -\boldsymbol{k} \cdot \boldsymbol{e}_{\hat{0}} \tag{5.4.6}$$

be the energy measured by a freely falling observer. Use the null condition $\boldsymbol{k} \cdot \boldsymbol{k} = 0$ to show that the two radial null directions can be written

$$\boldsymbol{k}_\pm = E_{\rm ff}\big(\boldsymbol{e}_{\hat{0}} \pm \boldsymbol{e}_{\hat{1}}\big). \tag{5.4.7}$$

The plus sign denotes locally outward propagation and the minus sign locally inward propagation. Note that the radial velocity of light measured by the freely falling observer $= -(\boldsymbol{k}_\pm \cdot \boldsymbol{e}_{\hat{1}})/(\boldsymbol{k}_\pm \cdot \boldsymbol{e}_{\hat{0}}) = \pm 1$. This is true for $r \geq 2M$ and $r < 2M$.

(d) **Non-radial light rays.** Let us include an additional orthonormal basis vector to the ones in (b): $\boldsymbol{e}_{\hat{2}} = \boldsymbol{e}_\theta^{\rm PG}/r$. At any event, consider a future-directed light ray making an angle α with the outward radial direction, as measured by the freely falling observer. By spherical symmetry, choose the ray to lie in the radial–θ plane. Show that its tangent vector can be written as

$$\boldsymbol{k} = E_{\rm ff}\big(\boldsymbol{e}_{\hat{0}} + \cos\alpha\, \boldsymbol{e}_{\hat{1}} + \sin\alpha\, \boldsymbol{e}_{\hat{2}}\big). \tag{5.4.8}$$

Express $\boldsymbol{k}$ in the PG co-ordinate basis and show that

$$\frac{{\rm d}T}{{\rm d}r} = \frac{1}{-v(r) + \cos\alpha}. \tag{5.4.9}$$

Argue that the radial slopes in the (r-T plane) of all future-directed light rays lie between those of the two radial null rays. (Note that α is the locally measured direction at an event; it need not remain constant along the ray.)

Exercise 5.4.2 : An observer at $0 < r < 2M$ emits two radial flashes, one locally outward and one locally inward. Both rays subsequently move toward smaller r. Can either flash meet the observer again before the observer reaches the singularity? Explain by comparing the null and timelike directions in a (T, r) spacetime diagram.

5.5 Collapse and the Disappearing Star

Refer to Fig. 5.2 for the next two sub sections.

Collapsing "Star": Consider a spherical "star" of mass M and with its surface at the radial co-ordinate $r = R > 2M$. The exterior ($r > R$) geometry is still Schwarzschild (*Birkoff's Theorem*), though the interior is not. The metric for the interior and exterior can be written compactly as (in inflalling PG co-ordinates)

$$\mathrm{d}s^2 = -\mathrm{d}T^2 + (\mathrm{d}r + v(T,r)\mathrm{d}T)^2 + r^2\mathrm{d}\Omega^2, \tag{5.5.1}$$

$$\text{with} \quad v(T,r) = \begin{cases} r\sqrt{\dfrac{2M}{R^3(T)}} & r \le R(T), \\ = \sqrt{\dfrac{2M}{r}} & r > R(T). \end{cases} \tag{5.5.2}$$

Here $r = R(T)$ is the radial co-ordinate of the star as a function of T. We have assumed that the spatial density of the star remains constant, but evolves with time as $\rho(T) = 3M/4\pi R(T)^3$. This is the Oppenheimer-Snyder (OS) collapse.

We will assume that the surface at R is falling inwards with $\boldsymbol{u}_R = \boldsymbol{e}_T - \sqrt{2M/R}\boldsymbol{e}_r$, where $\boldsymbol{e}_\alpha$ are co-ordinate basis vectors in the in falling PG co-ordinates. The equation for the stellar radius is

$$\frac{R(T)}{2M} = \left[\frac{3}{2}\frac{(T_* - T)}{2M}\right]^{2/3}, \tag{5.5.3}$$

which remains valid for $R(T) > 2M$ and also when $0 < R(T) < 2M$. This expression for $R(T)$ represents the case of a zero-pressure ball of "dust" which is marginally bound. We define a time T_h as the time when

$$R(T_\mathrm{h}) = 2M, \qquad T_\mathrm{h} = T_* - 4M/3. \tag{5.5.4}$$

Note that the time taken for $R(T)$ to go from $2M$ to 0, is $\Delta T = T_* - T_\mathrm{h} = 4M/3 \approx 7 \times 10^{-6}(M/M_\odot)\,\mathrm{s}$.

Light emitted from the star's surface: Let us use our understanding of null-geodesics to understand what happens to light. In particular, for $r > R(T) > 2M$, light emitted radially outward from the surface $r = R(T)$ can proceed to large r and reach $r \to \infty$ long as $R > 2M$, ie. $\mathrm{d}r/\mathrm{d}T > 0$ for such rays. When $R(T) = 2M$, the outward light ray has $\mathrm{d}r/\mathrm{d}T = 0$. Once $R(T) < 2M$, $\mathrm{d}r/\mathrm{d}T < 0$ for such rays emitted from the surface. So the star light from the surface stops being able to reach an observer at infinity once $R(T) < 2M$.

Exercise 5.5.1 : (a) For an observer riding on the star's surface, what is the speed of light measured by them for all $R(T) > 0$. (b) Suppose that light is emitted in regular proper time intervals of the infalling observer on the surface. Will the time-intervals between pulses received by a far away stationary observer be larger or smaller than this proper time interval of the in falling observer.

The distant observer's View: What does a distant observer see as $R(T)$ cross $2M$? How quickly does the star fade from view for a distant observer? Let us calculate the redshift of light emitted from the surface at a frequency $\omega_\star = -\boldsymbol{k}_+ \cdot \boldsymbol{u}_R$. We find that a stationary observer at $r = r_{\rm obs} \gg R(T)$ (and $r_{\rm obs} > 2M$), would see a redshift

$$\omega(r_{\rm obs}, T_{\rm obs}) \approx \omega_\star \left(1 - \sqrt{\frac{2M}{R(T)}}\right), \tag{5.5.5}$$

Where T is the emission time, and $T_{\rm obs}$ is the time of observation. They are linked by solving $dT/dr = 1/(-\sqrt{2M/r} + 1)$, with $T(r_{\rm obs}) = T_{\rm obs}$ and $T(R) = T$. As $R(T)$ approaches $2M$, we find

$$\omega(r_{\rm obs}, T_{\rm obs}) \propto \omega_\star e^{-T_{\rm obs}/4M}. \tag{5.5.6}$$

So for the distant observer, the frequency decreases exponentially, within an e-folding time $\Delta T_{\rm obs} \approx 2 \times 10^{-5} (M/M_\odot)\,{\rm s}$. While the star does not disappear suddenly, its brightness dims exponentially fast!

Exercise 5.5.2 : Derive the formulae for $\omega(r_{\rm obs}, T_{\rm obs})$ abov. First derive the general one for arbitrary $R(T) > 2M$ but assuming $r_{\rm obs} \gg R(T)$, and then consider the limit where $R(T)$ is approaching $2M$.

5.6 Horizon Formation

Now let us consider a hypothetical observer sitting at the center of the star at $r = 0$ while the star is collapsing. When the surface of the star is at $R(T) \gg 2M$, this observer starts sending out spherical light fronts. If the light fronts are emitted at a time $T_{\rm e}$ such that they reach the stellar surface while $R(T) > 2M$, they escape to infinity. There necessarily be a time $T_{\rm e} = T_{\rm b} < T_{\rm h}$ such that the front emitted at that time from the center will reach $r = R(T)$ precisely when $R(T) = 2M$. This front never goes beyond $r = 2M$. Every light front emitted for $T > T_{\rm b}$ never reaches $r = 2M$, though it can pass $R(T) < 2M$ for some T.

The Event Horizon: The spherical wavefront of the critical pulse, sent out at $T = T_{\rm b}$, generates a null hypersurface. Notice that the "radius" of this hypersurface $r_{\rm h}(T)$ for $T_{\rm b} < T < T_{\rm h}$ increases and then becomes constant at $r_{\rm h}(T \geq T_{\rm h}) = 2M$. This null hypersurface is the *event horizon*. This hypersurface is being generated from the time $T_{\rm b} < T_{\rm h}$ onwards, even before $R(T) = 2M$.

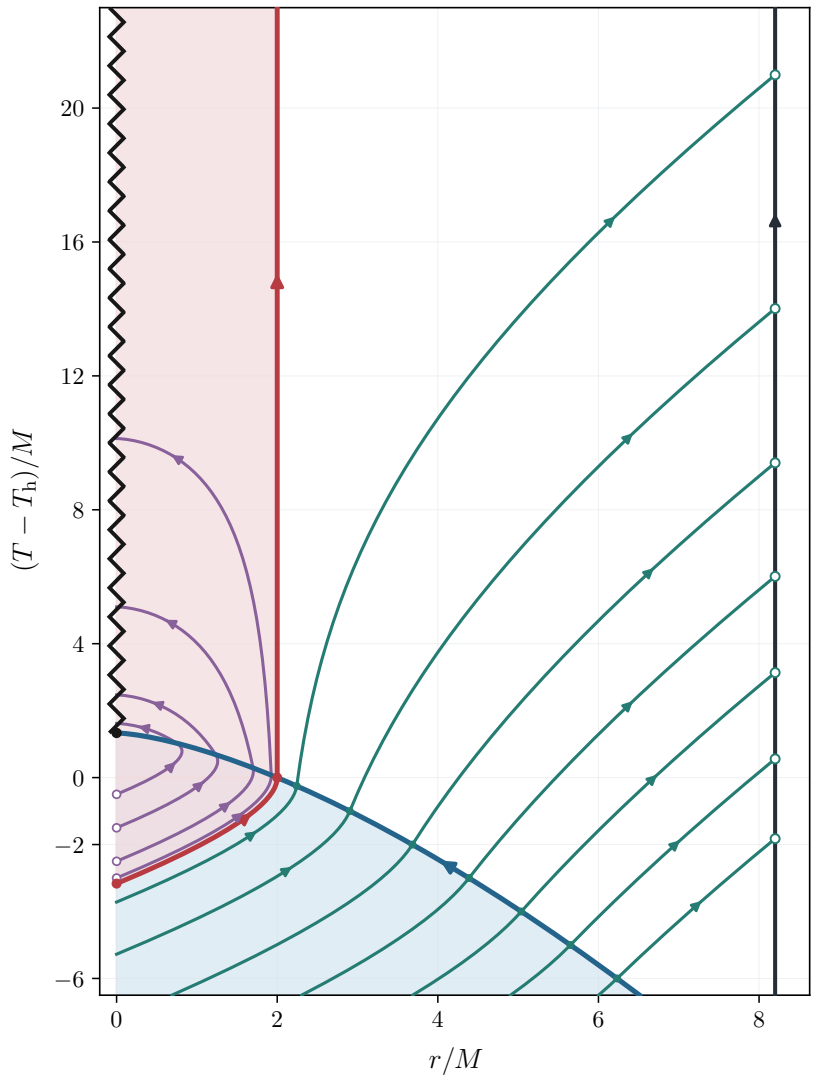


Figure 5.2: Outgoing radial light rays during marginally bound Oppenheimer–Snyder collapse, shown in infalling Painlevé–Gullstrand co-ordinates. The blue curve is the stellar surface $r = R(T)$; the red curve is the event horizon, which originates at the center at $T_\mathrm{b} = T_\mathrm{h} - 19M/6$ and meets the surface at $R(T_\mathrm{h}) = 2M$. Green rays escape to the distant stationary observer (black vertical line), whereas purple rays cannot escape and eventually reach $r = 0$. The blue shading denotes the stellar interior, the pink shading the region inside the event horizon, and the jagged black line the central singularity. Arrows indicate future-directed propagation.

Any event inside this null hypersurface (the event horizon) cannot communicate with an observer who remains outside the black hole. In particular, even before the stellar surface reaches $2M$, there are events inside the star from which no signal can ever escape to infinity. This is the remarkable feature of the event horizon: its "location" depends on the future evolution of spacetime.

If the hypothetical observer sending out the critical pulse from the center seems odd to you, you can also simply continue the outgoing radial light ray at the star's surface when $R(T) = 2M$ backwards and forwards in time. You will trace out the same null hypersurface.

Exercise 5.6.1 : Assume OS collapse.

(a) Show that the equation for the radial co-ordinate of the light fronts emitted at $r = 0$, $T = T_{\rm e}$ is

$$\mathrm{d}r/\mathrm{d}T = 1 - v(r, T). \tag{5.6.1}$$

Assume that the time of emission satisfies $T_{\rm b} \leq T_{\rm e} < T_*$.

(b) Show that their "turn-around" radius $r_{\rm turn}(T) = R(T)\sqrt{R(T)/2M}$. Note that it is smaller than $R(T)$ for $R(T) < 2M$.

(c) Verify that the solution for (a) is

$$r(T, T_{\rm e}) = 2R(T)\left[\sqrt{R(T_{\rm e})/2M} - \sqrt{R(T)/2M}\right], \tag{5.6.2}$$

(d) The critical light front emitted at $T_{\rm e} = T_{\rm b}$ reaches the stellar surface precisely when $R(T_{\rm h}) = 2M$. Use this condition to show that

$$R(T_{\rm b}) = \frac{9M}{2}, \qquad T_{\rm b} = T_* - \frac{9M}{2}. \tag{5.6.3}$$

Hence show that the event horizon inside the star is described by

$$\frac{r_{\rm h}(T)}{2M} = 3\left(\frac{3(T_* - T)}{4M}\right)^{2/3} - 3\frac{T_* - T}{2M}, \qquad T_{\rm b} \leq T \leq T_{\rm h}. \tag{5.6.4}$$

Verify that $r_{\rm h}(T_{\rm b}) = 0$, $r_{\rm h}(T_{\rm h}) = 2M$, and $\mathrm{d}r_{\rm h}/\mathrm{d}T = 0$ at $T = T_{\rm h}$. Putting it all together, the event horizon is described by

$$\frac{r_{\rm h}(T)}{2M} = \begin{cases} 3\left(\dfrac{3(T_* - T)}{4M}\right)^{2/3} - 3\dfrac{T_* - T}{2M} & T_{\rm b} \leq T < T_{\rm h} \\ 1 & T \geq T_{\rm h}. \end{cases} \tag{5.6.5}$$

Geometry of the event horizon and other hypersurfaces: The event horizon is a null hypersurface, and has a normal vector (which also lies in the hypersurface):

$$\boldsymbol{k}_{\rm h} = \boldsymbol{e}_T + [1 - v(T, r_{\rm h}(T))]\boldsymbol{e}_r. \tag{5.6.6}$$

Note that $\boldsymbol{k}_{\rm h} \cdot \boldsymbol{k}_{\rm h} = 0$. At $r_{\rm h}(T \geq T_{\rm h}) = 2M$, we have $\boldsymbol{k}_{\rm h} = \boldsymbol{e}_T$.

On the other hand, hypersurfaces of constant T are spacelike and Euclidean, both inside and outside the event horizon. Their future-directed unit normal vector is

$$\boldsymbol{u} = \boldsymbol{e}_T - v(T, r)\boldsymbol{e}_r. \tag{5.6.7}$$

This is precisely the four-velocity of the freely falling observers associated with the infalling PG coordinates, with $\boldsymbol{u} \cdot \boldsymbol{u} = -1$.

Finally, the $r = const.$ hypersurfaces are even more interesting. Their normal is

$$\boldsymbol{n} = v(T, r)\boldsymbol{e}_T + [1 - v^2(T, r)]\boldsymbol{e}_r. \tag{5.6.8}$$

The norm $\boldsymbol{n} \cdot \boldsymbol{n} = 1 - v^2$. These hypersurfaces are timelike for $v < 1$, null for $v = 1$ and spacelike for $v > 1$. Note that after $T > T_*$, $v(T, r) = \sqrt{2M/r}$, and $\boldsymbol{n} \cdot \boldsymbol{n} < 0$ for $r < 2M$ – ie. all constant r hypersurfaces are spacelike.

The Singularity: As the star collapses, for our OS collapse, its density increases as

$$\rho(T) = \frac{3M}{4\pi R^3(T)} = \frac{1}{6\pi(T_* - T)^2}. \tag{5.6.9}$$

which diverges as $T \to T_*$. All part of the star reach the $r \to 0$ at the same time T_*. This is a genuine "curvature" singularity, unlike the apparent singularity of Schwarzschild co-ordinates at $r = 2M$.

After the collapse, the spacetime for $r > 0$ is Schwarzschild. Recall that inside $r = 2M$, hypersurfaces of constant r are spacelike. In the limit $r \to 0$, they approach the singular boundary. We therefore call $r = 0$ a *spacelike singularity.*

The singularity is not an object sitting at the center of space that an observer can choose to avoid. For an observer inside the black hole, the singularity lies inevitably in their future. This can also be seen from the fact that all light emitted from within the event horizon $r < r_h(T)$ also ends up at the the singularity.

5.7 Additional Exercises

These are adapted conceptually from Hartle, Chapter 12, Problems 12.5, 12.8, 12.14. For these problems assume the Schwarzchild spacetime, with metric given in the in going PG co-ordinates $\mathrm{d}s^2 = -dT^2 + (dr + v(r)dT)^2 + r^2\mathrm{d}\Omega^2$, with $v(r) = \sqrt{2M/r}$.

Exercise 5.7.1 In the Schwarzschild spacetime, an observer is released from rest relative to a stationary observer at $r_0 = 10M$ and falls freely toward $r = 0$.

(a) Since the metric is independent of T, the energy per unit mass $e = -\boldsymbol{u} \cdot \boldsymbol{e}_T$ is conserved along the geodesic. Express e in terms of $\mathrm{d}T/\mathrm{d}\tau$ and $\mathrm{d}r/\mathrm{d}\tau$ where τ is the observer's proper time. Use $\boldsymbol{u} \cdot \boldsymbol{u} = -1$ to show that

$$\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 = e^2 - f(r). \tag{5.7.1}$$

(b) Determine e from the initial condition, and calculate the total proper time between release at $r_0 = 10M$ and arrival at $r = 0$. Express your answer in units of M.

Exercise 5.7.2 An observer freely falling from rest at infinity crosses $r = 2M$ at PG time $T = 0$, reaching $r = 0$ at $T = 4M/3$. A stationary source outside the horizon, at $r = r_{\mathrm{e}} > 2M$, sends inward-directed light pulses.

(a) Can the infalling observer receive a pulse emitted from outside the horizon *after* crossing $r = 2M$? Sketch the observer's worldline and suitable ingoing null rays in a (T, r) diagram.

(b) Can the infalling observer receive signals from *every* event outside the black hole? Explain using the finite proper time remaining before the singularity.

Exercise 5.7.3 Consider any future-directed timelike worldline after it crosses $r = 2M$ in the Schwarzschild black-hole spacetime. Unlike the freely falling observers used to define PG time, this observer may accelerate and move in nonradial directions.

(a) Inside the horizon, r decreases along every future-directed timelike worldline. Use r as a parameter, and show from the PG metric that

$$\left(\frac{\mathrm{d}\tau}{\mathrm{d}r}\right)^2 \leq \frac{1}{v^2(r) - 1} = \frac{r}{2M - r}. \tag{5.7.2}$$

Hint: Complete the square in $\mathrm{d}T/\mathrm{d}r$ and use the fact that angular motion can only reduce the proper time.

(b) Integrate the bound to show that the proper time between horizon crossing and the singularity cannot exceed πM. Compare with the $4M/3$ experienced by an observer falling freely from rest at infinity.
